

Risk Aversion Reversals^{*}

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Abstract

We characterize when risk aversion reversals—situations in which a more risk-averse individual chooses a riskier asset more frequently—arise in weak utility models of binary stochastic choice, a non-parametric class that encompasses virtually all models of stochastic choice used in practice. Reversals necessarily arise if and only if the ratio of the second derivatives of the individuals’ utilities is unbounded. Reversals are robust to arbitrary heterogeneity and choice of utility representation. Our condition is violated by the most widely used utility specifications including CARA, CRRA, and utilities without constant risk aversion such as expo-power. Under these specifications, reversals are pervasive, occurring for empirically relevant risk coefficients and lottery choices. We thus propose two parametric utility families that do not lead to reversals, offering well-behaved alternatives for stochastic choice. Finally, we analyze reversals in multinomial settings under two frameworks: the additive perturbed utility model and the random parameter model. We characterize the class of utilities immune to reversals, which includes our proposed families, but excludes CARA and CRRA families.

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1 Introduction

A precise understanding of how humans evaluate risk is central to economics. Since choices are inherently noisy, analysts deploy stochastic choice models to recover individuals’ risk preferences and noise from choice data. The most widely used stochastic choice models for this purpose are *Fechnerian models*, a non-parametric class that includes the logit and probit models. In practice, the analyst chooses a functional form for utility and a functional form for noise which maps utility differences to choice probabilities (e.g., Harrison, List, and Towe, 2007; Von Gaudecker, Van Soest, and Wengström, 2011; Holzmeister and Stefan, 2021).

However, predictions from these models under CARA or CRRA—the most commonly used utility functions—suffer from a well-known problem. When two individuals differ in their risk-aversion parameters but face the same noise, the more risk-averse individual exhibits more risk-seeking behavior in some choices (Wilcox, 2008, 2011; Blavatsky, 2011; Apesteguia and Ballester, 2018). In response to these critiques, subsequent work has argued that non-monotonicity is particularly worrisome under homoskedastic noise and has advocated allowing heteroskedasticity to better calibrate noise to utility (Barseghyan, Molinari, O’Donoghue, and Teitelbaum, 2018; O’Donoghue and Somerville, 2024; Keffert and Schweizer, 2024). Moreover, it is common to model heterogeneity in both preferences and noise and to estimate them jointly (e.g., Hey and Orme, 1994; Von Gaudecker, Van Soest, and Wengström, 2011; Meissner, Gassmann, Faure, and Schleich, 2023).

In this paper, we show that the problems noted for Fechnerian models are far more pervasive than previously recognized: they persist even when individuals differ arbitrarily in their Fechnerian noise, when noise varies across menus, and when noise takes a more general, non-Fechnerian form. That is, under CARA and CRRA utilities, none of these generalizations can eliminate the counterintuitive predictions. We then characterize which utility functions, beyond CARA and CRRA, give rise to risk-aversion reversals and advocate for two parametric utility families that do not, under various noise specifications, exhibit reversals: a more risk-averse individual always chooses safer options more frequently. Beyond its intuitive appeal, this comparative static aligns with empirical findings (Bruner, 2017). Finally, we discuss a generalization of Fechnerian models to multinomial choice and show that our proposed utility families do not exhibit reversals even in this more general setting.

To build intuition, we begin with the baseline Fechnerian framework, in which each individual is associated with a Bernoulli utility function u and a strictly increasing function F . We refer to F as a Fechnerian noise structure, where

$$F(U(X) - U(Y))$$

is the probability that X is chosen over Y , and $U(X) = \mathbb{E}[u(X)]$. The most popular Fechnerian noise structures are CDFs of normal and logistic distributions, corresponding to probit and logit models. Indeed, any additive random utility models with i.i.d. shocks fit the Fechnerian framework.¹

Our first result is that, for some pairwise choices between a safe and a risky lottery,² a more risk-averse individual chooses the risky option more often than a less risk-averse individual, even when the two differ arbitrarily in their Fechnerian noise structures (Proposition 1). Thus, the problems identified for the homoskedastic noise model cannot be resolved by jointly estimating risk and noise, even when noise is estimated non-parametrically.

For example, consider an analyst who observes Anne and Bob choosing from three distinct menus—each menu consists of a safe lottery—a sure payoff of \$8, \$10, or \$14—and a 50–50 risky lottery paying plus or minus \$4, \$6, or \$8 from the sure amount. The choice probabilities for the safe lotteries are shown in the second and third columns of Table 1. Assuming CRRA expected utilities for their risk preferences and normal distributions for their noise structures, Anne is estimated to be more risk-averse and experience less noise than Bob. Indeed, the choice probabilities in Table 1 correspond to Anne having a CRRA coefficient of 0.8 and noise variance of 0.5, compared to Bob’s 0.3 and 1, respectively.³

¹Technically, for F to be strictly increasing, the difference in these shocks must have full support.

²We say that lottery X is safer than lottery Y if X dominates Y in the concave order. Equivalently, every expected utility maximizer with a concave utility function must prefer X to Y .

³I.e., the analyst deploys the probit model for Anne and Bob

$$F_A(U_A(X) - U_A(Y)) \quad \text{and} \quad F_B(U_B(X) - U_B(Y)),$$

respectively, where U_A and U_B are CRRA expected utility functions with relative risk aversions a and b , and F_A and F_B are the CDFs of normal distributions with zero means and respective variances σ_A^2 and σ_B^2 .

Base Lottery (S vs R)	Probability of S		Probability of $3S$	
	Anne	Bob	Anne	Bob
8 vs 4, 12	0.60	0.57	0.62	0.65
10 vs 4, 16	0.65	0.62	0.69	0.74
14 vs 6, 22	0.64	0.63	0.68	0.77

Table 1: CRRA Choice Probabilities for S with normal errors.

In the last two columns of Table 1, we report the choice probabilities of safe options whose outcomes are tripled from the base lotteries under the estimated parameters. Note that when the stakes are increased, Bob chooses safer options more frequently than Anne from all menus. Thus, despite Anne being estimated as more risk-averse and more precise, the model predicts that she will behave more risk-seeking than Bob once the stakes are scaled up.

This reversal pattern does not hinge on these specific CRRA risk coefficients and noise structures, nor on the specific choice of lotteries. Indeed, given lotteries X and Y with X safer than Y , we show how to construct lotteries X' and Y' offering higher potential rewards, such that X' remains safer than Y' yet leads to a reversal. For CARA utilities, reversals occur after shifting the lotteries by the same sufficiently large outcome (Proposition 3). Thus, CARA and CRRA Fechnerian models with arbitrary noise structures make the perverse and systematic prediction that those most inclined toward choosing safer options when stakes are small must become the least inclined when stakes are larger.

Assuming that this reversal pattern does not always bear out in reality—and indeed Bruner (2017) finds that it does not—our results indicate that the estimates yielded by these models will be sensitive to the level of stakes that are analyzed, making them unreliable.⁴ Indeed, if the analyst instead observed the choice probabilities of the scaled lotteries in Table 1, the analyst would estimate that Bob is more risk-tolerant than Anne, even if he exhibited more risk-averse behavior. As illustrated in that table, the stake increases are often modest, indicating that such reversals

⁴This echoes the critique of expected utility in Rabin (2000), which also reveals paradoxes when stakes are increased. While Rabin’s result is about the curvature of utility functions in deterministic models, our result is about the interaction between noise structures and CARA/CRRA utilities.

can emerge even in low-stakes experimental settings. Moreover, these models give counterintuitive, and perhaps counterfactual, out-of-sample predictions for choices made at larger or smaller stakes.

We next show that the paradoxical reversals of CARA and CRRA Fechnerian models are not artifacts of overly restrictive Fechnerian noise structures. To establish this, we explore *weak expected utility* (WEU) models. A WEU model consists of a vNM utility function U , a Fechnerian noise structure F and a menu-dependent scale parameter $\sigma: \mathcal{L} \times \mathcal{L} \rightarrow (0, \infty)$ such that the probability of choosing X over Y is given by

$$F\left(\frac{U(X) - U(Y)}{\sigma(X, Y)}\right).^5$$

WEU models generalize the standard Fechnerian framework by allowing choice probabilities to depend on factors beyond strict utility differences. An important subclass of WEU models arises in the additive random utility framework, when the random shocks to utility that are associated with each lottery are independent but not identically distributed. Economically, this accommodates context-dependent noise; for example, large-stakes lotteries may induce increased attention, resulting in less variable shocks.

Without any restrictions on how noise varies across menus, the only implication of WEU models is that individuals will choose their preferred lottery more than half the time. Thus, to derive meaningful comparisons between individuals with different risk attitudes, we must impose additional structure. Our next results (Theorem 1 and Corollary 1) show that, under a mild continuity requirement on the menu-dependent scale parameters, WEU models based on CARA or CRRA utilities are still guaranteed to produce reversals. Thus, even when allowing for arbitrary noise that varies across both individuals and menus, the CARA and CRRA expected utility functional forms lead to untenable predictions.

One might suspect, at this point, that these reversals arise because F and σ are applied to vNM utilities, which are just representations of preferences and carry no cardinal significance. A natural alternative would be to apply F and σ to certainty equivalents $u^{-1}(\mathbb{E}[u(X)])$, or more generally to $f(\mathbb{E}[u(X)])$ for some strictly increasing transformation f . However, Corollary 2 establishes that this

⁵We require $\sigma(X, Y) = \sigma(Y, X)$ so that the probabilities sum up to 1.

transformation does not resolve the problem. Ultimately, CARA and CRRA utilities prove to be fundamentally incompatible with stochastic choice consistency.

We next turn our attention toward the characterization of families of Bernoulli utilities that interact well with noise. Our main result is that when u_A is more risk-averse than u_B , there exist reversals for every weak utility model if and only if the ratio

$$u_B''(x)/u_A''(x)$$

is unbounded (Theorem 1). Indeed, for CARA and CRRA utilities, this condition holds, leading to reversals in every model of noise we have considered. Moreover, this condition is met by a class of utility functions that generalizes CARA and CRRA (Corollary 3).

We thus advocate for two families of utility functions for which there are no reversals under homoskedastic noise (see Section 5.1). These families are obtained by taking a linear combination of two distinct CARA (CRRA) utilities and are parameterized by the coefficient on the more risk-averse utility function. For example, for any $0 \leq r < s$, let crra_r and crra_s be CRRA Bernoulli utility functions with respective relative risk coefficients r and s . Summing $\text{crra}_r + a \cdot \text{crra}_s$ yields a family of utility functions parameterized by $a \geq 0$, with greater a corresponding to greater risk aversion. These simple families of utility functions offer well-behaved alternatives for empirical analysis. We discuss the properties of these utility families and the interpretation of the parameter a in Appendix F.

We next study reversals in the more general multinomial-choice setting. We introduce two competing definitions that extend the notion of reversals from the binary-choice setting to multinomial choice. We focus on the *additive perturbed utility* (APU) model of Fudenberg, Iijima, and Strzalecki (2015), a multinomial extension of the Fechnerian model. Our main result is that the utility families that avoid reversals in the binary setting also avoid both notions of reversals in the multinomial setting (Theorem 2).

Finally, we show that the *random parameter* models considered by Apesteguia and Ballester (2018) lead to one notion of reversal in the multinomial setting, but not the other (Proposition 6). We conclude by characterizing utility families that do not suffer from any reversals (Theorem 3).

1.1 Related Literature

We contribute to a large body of literature on risky choices in the presence of noise (e.g., Becker, DeGroot, and Marschak, 1963; Harless and Camerer, 1994; Hey and Orme, 1994; Loomes, Moffatt, and Sugden, 2002; Blavatskyy, 2007). See Wilcox (2021) for a recent survey. In particular, we study Fechnerian noise structures and their generalizations. Axiomatic investigations of these models have been undertaken by Debreu (1958), Tversky and Russo (1969), and Fudenberg, Iijima, and Strzalecki (2015), among others. See Strzalecki (2025) for an extensive review.

The papers most closely related to ours are Wilcox (2011) and Apesteguia and Ballester (2018), which demonstrate that CARA and CRRA utilities coupled with identical Fechnerian noise are problematic: a more risk-averse individual exhibits more risk-seeking behavior in some pairwise choices. Wilcox (2011) shows that other parameterizations of CARA and CRRA utilities also lead to reversals. This corresponds to two individuals having noise structures that are related by a scale factor. We study this special case in Proposition 2. Our work builds on their insights in several ways. First, we allow for heterogeneous noise structures and show that the paradox persists without any parametric assumptions on how noise differs across individuals (Proposition 1). Second, we show that the reversals occur under WEU models, in which noise varies across menus (Corollary 1).

WEU models recognize that some comparisons may involve greater cognitive effort or uncertainty. Menu-dependent noise in risky choice has been extensively studied. Hey (1995), Buschena and Zilberman (2000), and Loomes (2005) model Fechnerian noise that depends on, for example, value differences between lotteries and question difficulty. More recently, He and Natenzon (2024) and Shubatt and Yang (2024) characterize Fechnerian noise that depends on a measure of distance between alternatives.

Finally, we characterize which families of utility functions, beyond CARA and CRRA, suffer from these reversals (Theorem 1) and propose alternative parametric families of utilities that do not suffer from the problem. This is in contrast with the solutions proposed by Wilcox (2011) and Apesteguia and Ballester (2018) who retain CARA and CRRA utilities.

Wilcox (2011) considers a WEU model and shows it is monotone for lotteries over three fixed outcomes. Nevertheless, Apesteguia and Ballester (2018) show that

this model is no longer monotone for more than three outcomes and propose instead the random parameter model, where each individual is associated with a distribution over CARA or CRRA preferences and chooses probabilistically as if they draw a random preference from their distribution. Random parameter models do not give rise to reversals in binary choice, but they are based on a different foundation than traditional models of noise. In particular, in random parameter models there is no longer a core preference over alternatives such that an individual usually chooses the more preferred outcome. This is in contrast with our proposal to use alternative vNM utility functions which maintains the idea that individuals choose according to a core preference under noise. Moreover, we show that the random parameter models advocated for by Apesteguia and Ballester (2018) can lead to what we call *compound reversals* when applied to menus with three or more alternatives. This is in contrast with the models we advocate for, which do not generate compound reversals in multinomial choice contexts.

Another paper that studies risk aversion in stochastic choice is Blavatskyy (2011), which examines when one individual is always more likely to choose a sure thing over a lottery than another individual. A main result of that paper is that this notion of stochastic comparative risk aversion is too strong to be compatible with any Fechnerian expected utility models.

We compare the riskiness of lotteries using the concave order of Rothschild and Stiglitz (1970). In the deterministic setting, Kihlstrom, Romer, and Williams (1981) and Ross (1981) have noted that insurance and risk premia are not increasing in the Arrow-Pratt order. Ross (1981) thus proposes a stronger notion of comparative risk preferences than Arrow-Pratt that leads to monotonicity. Our characterization result (Theorem 1) shows that this stronger notion implies our condition, and thus is sufficient to eliminate reversals under some noise structures.

Finally, we contribute to a literature examining failures of expected utility theory in predicting choices across varying ranges of stakes. The most prominent critique in this area is by Rabin (2000), who demonstrates that within the expected utility framework, even modest levels of risk aversion exhibited over small-stakes, imply absurd, counterfactual levels of risk aversion for large stakes. While this result holds for deterministic choice, Propositions 2 and 3 echo Rabin (2000) in the stochastic setting: out of sample predictions made from small-stakes observations are counterintuitive.

2 Preliminaries

We denote by $\mathcal{L}$ the set of all bounded real random variables defined over a non-atomic probability space $(\Omega, \Sigma, \mathbb{P})$. We use the term *lotteries* to refer to elements of $\mathcal{L}$. A stochastic choice rule is a map $\rho: \mathcal{L} \times \mathcal{L} \rightarrow \mathbb{R}$ where $\rho(X, Y) + \rho(Y, X) = 1$ and $\rho(X, Y)$ is the probability of choosing X over Y . A vNM *utility function* is a map $U: \mathcal{L} \rightarrow \mathbb{R}$ given by $U(X) = \mathbb{E}[u(X)]$ for some strictly increasing, concave, and continuous Bernoulli utility $u: \mathbb{R} \rightarrow \mathbb{R}$.

We consider an individual that evaluates lotteries according to a vNM utility U but faces noise and chooses stochastically according to ρ . We think of ρ as a noisy expression of the underlying utility U . For example, an individual with utility U whose noise takes a logit form has choice probabilities

$$\rho(X, Y) = \frac{e^{\beta U(X)}}{e^{\beta U(X)} + e^{\beta U(Y)}},$$

for some $\beta > 0$. When β is fixed across all menus, ρ belongs to the class of Fechnerian models, where choice probabilities are a function of utility differences. We discuss Fechnerian models in depth in Section 3. More generally, the logit scale parameter β may depend on (X, Y) . We study these general models in Section 4.

We say that X dominates Y in the *concave order*, denoted $X \geq_c Y$ if $\mathbb{E}[g(X)] \geq \mathbb{E}[g(Y)]$ for all concave functions $g: \mathbb{R} \rightarrow \mathbb{R}$. We denote the strict part of $\geq_c$ by $>_c$. Recall that $X \geq_c Y$ if and only if Y is a mean-preserving spread of X . If $X \geq_c Y$, then $U(X) \geq U(Y)$ for all vNM utilities U .⁶ We will often use the variables S and R when referring to lotteries where $S >_c R$, in order to highlight that S is *safer* while R is *riskier*.⁷ Finally, we denote by C_+^k the set of k -times continuously differentiable functions defined on an open interval of $\mathbb{R}$ with strictly positive k th derivatives.

2.1 Stochastic Comparative Risk Aversion

We are interested in comparing the choices of individual A with those of B . Each individual i is associated with a utility function U_i and stochastic choice model ρ_i . We will often refer to agents and their associated stochastic choice models

⁶Since we defined vNM utilities as expectations of concave Bernoulli utilities, they are increasing in $\geq_c$.

⁷While we focus on lotteries that can be ranked in the concave order, analogous results hold throughout for lotteries comparable by second-order stochastic dominance. See Section 5.1.

interchangeably. We say A is *weakly more risk-averse than* B if $U_A(X) \geq U_A(y)$ implies $U_B(X) \geq U_B(y)$ for any $X \in \mathcal{L}$ and degenerate lottery y .⁸ We say A is *more risk-averse than* B if A is weakly more risk-averse than B and B is not weakly more risk-averse than A .

Definition 1. When A is more risk-averse than B , we say that there is a reversal if there are lotteries $S >_c R$, where

$$\rho_A(S, R) < \rho_B(S, R).$$

Moreover, we say that S and R generate a reversal.

This means that A , the more risk-averse individual, exhibits more risk-tolerant behavior by choosing the safer lottery S less frequently than B . A 's behavior can also be seen as more risk-tolerant in the following sense: Let s_A and s_B denote A 's and B 's respective choice probabilities of S . The ex-ante lottery that A receives is the compound lottery resulting in S with probability s_A and in R with probability $1 - s_A$, and likewise for B . Since $s_A < s_B$, A 's compound lottery is riskier than B 's, so A evidently makes riskier choices than B . These two ways of interpreting reversals in binary choice naturally lead to different extensions of the notion of reversal to multinomial choice; see Section 6.

In the following sections, we show that many models of noise lead to reversals under the Constant Absolute Risk Aversion (CARA) and Constant Relative Risk Aversion (CRRA) utility specifications, the most commonly used parametric families in both theory and practice. CARA and CRRA utilities are defined by $\text{CARA}_a(X) = \mathbb{E}[\text{cara}_a(X)]$ and $\text{CRRA}_a(X) = \mathbb{E}[\text{crra}_a(X)]$, where $a \geq 0$ is the coefficient of absolute/relative risk aversion and

$$\text{cara}_a(x) = \begin{cases} x & a = 0 \\ \frac{1 - e^{-ax}}{a} & a > 0, \end{cases} \quad \text{crra}_a(x) = \begin{cases} \frac{x^{1-a} - 1}{1-a} & a \neq 1 \\ \ln(x) & a = 1. \end{cases}$$

⁸This is Yaari (1969)'s notion of comparative risk, which coincides with the Arrow-Pratt notion of comparative risk since U_A and U_B are vNM utility functions. I.e., U_A being more risk-averse than U_B is equivalent to the concavity of $u_A \circ u_B^{-1}$, when u_A and u_B are the corresponding Bernoulli utility functions to U_A and U_B . Indeed, CARA and CRRA utilities are totally ordered by this order.

The coefficients reflect varying levels of risk aversion, with higher coefficients corresponding to greater aversion to risk.⁹ To ensure that $\text{CRRA}_a(X)$ is well defined, we restrict our attention to X that is non-negative and bounded away from 0 whenever referring to CRRA.

3 Fechnerian Models

We first consider models of the form

$$\rho(X, Y) = F(U(X) - U(Y)), \quad (1)$$

where $F: \mathbb{R} \rightarrow [0, 1]$ is a C_+^1 function satisfying $F(t) + F(-t) = 1$. We denote by $\mathcal{F}$ the set of all such functions and refer to a model of the form (U, F) , where U is a vNM utility and $F \in \mathcal{F}$, as a *Fechnerian expected utility* (FEU) model.¹⁰ In these models, the larger the difference between $U(X)$ and $U(Y)$, the higher the probability that X is chosen.

FEU models are among the most widely used stochastic choice models (Becker, DeGroot, and Marschak, 1963; Loomes, Moffatt, and Sugden, 2002). An important subclass of FEU models yields choice probabilities

$$\mathbb{P}(U(X) + \varepsilon \geq U(Y) + \varepsilon') = F(U(X) - U(Y)), \quad (2)$$

where ε and ε' are i.i.d. continuous random variables and F is the CDF of $\varepsilon - \varepsilon'$. This subclass includes many widely used discrete choice models such as the logit and probit models, where the random shocks are Gumbel and Gaussian, respectively.

In this section, we focus on FEU models and show that the CARA and CRRA specifications display a paradox: they predict that a more risk-averse individual will sometimes choose a riskier asset more frequently than a more risk-tolerant individual under any Fechnerian noise specifications.

⁹Note that there are many possible ways of parameterizing CARA and CRRA preferences, since applying a positive affine transformation to each Bernoulli utility does not change the underlying preference/coefficient. Nevertheless, we will show that our results hold for every possible parameterization of these families.

¹⁰The assumptions that F must be continuously differentiable with positive derivative can be relaxed without affecting our results, provided these assumptions hold at 0. Note that we do not assume that $\lim_{t \rightarrow \infty} F(t) = 1$, nor that $\lim_{t \rightarrow -\infty} F(t) = 0$, although this will be the case for the examples we study.

Proposition 1. *For any $F_A, F_B \in \mathcal{F}$ and distinct CARA (CRRA) utilities U_A, U_B , there exist reversals.*

Since CARA (CRRA) preferences are totally ordered by risk-aversion, without loss of generality suppose A is more risk-averse. Proposition 1 says that there nevertheless exist lotteries $S >_c R$ such that $\rho_A(S, R) = F_A(U_A(S) - U_A(R)) < F_B(U_B(S) - U_B(R)) = \rho_B(S, R)$.

Proposition 1 is an impossibility result for the commonly used CARA/CRRA-expected-utility Fechnerian models since reversals arise under arbitrary pairs of noise structures and risk coefficients. In the case of $F_A = F_B$, i.e., when individuals face the same noise structures, it was shown by Wilcox (2011) and Apesteguia and Ballester (2018) that the probability of choosing the safer option from a pair is non-monotone in the CARA/CRRA coefficients.¹¹ In practice, however, it is common to model heterogeneity in both preferences and noise and to estimate them jointly (Hey and Orme, 1994; Von Gaudecker et al., 2011). Indeed, subsequent work has argued that non-monotonicity is particularly worrisome under homoskedastic noise and has advocated allowing heteroskedasticity for better calibration of each individual’s noise to their utility scale (Barseghyan et al., 2018; O’Donoghue and Somerville, 2024; Keffert and Schweizer, 2024).

Yet, perhaps surprisingly, Proposition 1 demonstrates that non-monotonicity persists even when noise structures vary arbitrarily across individuals. Thus, no matter how we jointly estimate risk coefficients and noise structures for two individuals, the resulting estimated models (CARA_a, F_A) and (CARA_b, F_B) will predict reversals. Because reversals are inherent to these models, we must be cautious when using these estimates to rank individuals by risk aversion.

Note that choice probabilities are not invariant to reparameterizations of CARA and CRRA utilities. For example, consider the model (U, F) , where U is a CARA or CRRA utility and $F \in \mathcal{F}$. Let $V(X) = \frac{1}{c}U(X) + d$ for some $c > 0$. While V represents the same preference as U , replacing U with V has the same effect on choice probabilities as changing the noise structure from F to G which is defined by $G(t) = F(t/c)$. However, since G is also a member of $\mathcal{F}$, it follows from Proposition 1 that reparameterizations of the CARA or CRRA utility family cannot resolve the non-monotonicity.

¹¹When $F_A = F_B$, Proposition 1 follows from Proposition 3 of Apesteguia and Ballester (2018).

3.1 Scale-Family Heteroskedasticity

To develop intuition for Proposition 1, we first consider individuals with distinct CARA or CRRA utilities U_A and U_B with respective coefficients a and b , where $a > b$ and whose noise structures are related by a scale factor. That is, $F_A(t) = F(t/\sigma_A)$ and $F_B(t) = F(t/\sigma_B)$ for some $\sigma_A, \sigma_B > 0$ and common $F \in \mathcal{F}$.¹² The choice probabilities of S over R are then given by

$$F\left(\frac{U_A(S) - U_A(R)}{\sigma_A}\right) \quad (3)$$

$$F\left(\frac{U_B(S) - U_B(R)}{\sigma_B}\right). \quad (4)$$

Since F is strictly increasing, the pair (S, R) generates a reversal (i.e., (4) is strictly larger than (3)) if and only if

$$\frac{U_B(S) - U_B(R)}{U_A(S) - U_A(R)} > \frac{\sigma_B}{\sigma_A}. \quad (5)$$

Note that $\frac{\sigma_B}{\sigma_A}$, which is the relative noise level, does not depend on the lotteries. Given this, the next lemma immediately implies that to generate a reversal, we can start with any $S >_c R$, and get a reversal by adding to both a large enough constant x (in the CARA case) or multiplying both by a large enough constant (in the CRRA case).

Lemma 1. *For any $S >_c R$, $a, b > 0$, and all $x \in \mathbb{R}$ and $k > 0$,*

$$1. \frac{\text{CARA}_b(S + x) - \text{CARA}_b(R + x)}{\text{CARA}_a(S + x) - \text{CARA}_a(R + x)} = C_1 \cdot e^{(a-b)x}$$

$$2. \frac{\text{CRRA}_b(k \cdot S) - \text{CRRA}_b(k \cdot R)}{\text{CRRA}_a(k \cdot S) - \text{CRRA}_a(k \cdot R)} = C_2 \cdot k^{a-b},$$

for some positive constants C_1 and C_2 .

Lemma 1 is proved in Appendix A. Note that when $a > b$, the ratios are strictly increasing in x and k and tend to infinity. Moreover, these properties hold for any $S >_c R$. We thus have the following result which strengthens the existence of reversal pairs established in Proposition 1 when noise structures are related by a scale factor.

¹²This is the case, for example, when F_A and F_B are both normal or both logistic CDFs.

Proposition 2. *Let $\sigma_A, \sigma_B > 0$ and $F \in \mathcal{F}$. Let $F_A(t) = F(\frac{t}{\sigma_A})$, $F_B(t) = F(\frac{t}{\sigma_B})$, and $S >_c R$. Let U_A, U_B be distinct CARA utilities and let V_A, V_B be distinct CRRA utilities. Then there exist unique x_0 and k_0 such that*

1. *$S + x$ and $R + x$ generate a reversal for (U_A, F_A) and (U_B, F_B) if and only if $x > x_0$;*
2. *kS and kR generate a reversal for (V_A, F_A) and (V_B, F_B) if and only if $k > k_0$.*

Note that the critical values x_0 and k_0 given in Proposition 2 depend on the noise scale parameters σ_A, σ_B , risk coefficients, and the baseline lotteries. The values can be derived from (5) and the positive constants C_1 and C_2 given in Lemma 1. I.e., critical values x_0 and k_0 solve for $C_1 \cdot e^{(a-b)x} = \frac{\sigma_B}{\sigma_A}$ and $C_2 \cdot k^{(a-b)} = \frac{\sigma_B}{\sigma_A}$.¹³ For example, when the CRRA coefficients a and b are 0.8 and 0.3, respectively, and the noise variances are $\sigma_A = 0.5$ and $\sigma_B = 1$, the critical values that generate reversals for lotteries given in Table 1 are all less than 3, which illustrates that reversals often occur under empirically relevant risk coefficients and lotteries.

Proposition 2 highlights two problems with the CARA and CRRA Fechnerian models. First, there is a disconnect between the standard Arrow-Pratt notion of comparative risk-aversion developed in the deterministic framework with the resulting stochastic choice predictions. Namely, more risk-averse individuals are not more likely to make risk-averse choices. This counterintuitive prediction is contradicted by empirical evidence that risk aversion is negatively correlated with decision noise, as measured by the frequency with which individuals choose the concave-order dominated option (Bruner, 2017). In terms of estimation, this means that even if Bob chooses safer lotteries more frequently than Anne, we may conclude that Bob is more risk-tolerant than Anne.¹⁴ This is the case, for example, if we only observe the choice probabilities of the scaled lotteries in Table 1.

Second, these models make the systematic prediction that those most inclined to choose safer options when stakes are small must become the least inclined once the

¹³Exact expressions for C_1 and C_2 are given in (14). We obtain

$$x_0 = \frac{1}{a-b} \ln \left(\frac{\sigma_B}{\sigma_A} \frac{b}{a} \left(\frac{\mathbb{E}[e^{-aR}] - \mathbb{E}[e^{-aS}]}{\mathbb{E}[e^{-bR}] - \mathbb{E}[e^{-bS}]} \right) \right), \quad k_0 = \left(\frac{\sigma_B}{\sigma_A} \frac{1-b}{1-a} \left(\frac{\mathbb{E}[S^{1-a}] - \mathbb{E}[R^{1-a}]}{\mathbb{E}[S^{1-b}] - \mathbb{E}[R^{1-b}]} \right) \right)^{\frac{1}{a-b}}.$$

¹⁴This holds under any consistent estimator.

stakes grow modestly larger. Even without a specific model of risk aversion in mind, this prediction is implausible.

3.2 Arbitrary Fechnerian Noise Structures

So far we have considered the special case that F_A and F_B are related by a scale factor. To prove the more general case considered in Proposition 1, we establish the following lemma, which shows that the properties of CARA and CRRA utilities given in Lemma 1 give rise to non-monotonicity for any $F_A, F_B \in \mathcal{F}$.

Lemma 2. *Let $F_A, F_B \in \mathcal{F}$. Let U_A and U_B be vNM utility functions with U_A more risk-averse. Suppose that for each $M \in \mathbb{R}$ there exist lotteries $S >_c R$ such that*

$$\frac{U_B(S) - U_B(R)}{U_A(S) - U_A(R)} \geq M. \quad (6)$$

Then there are reversals for (U_A, F_A) and (U_B, F_B) .

Lemma 2 identifies a sufficient condition on the vNM utility functions to yield the negativity result in Proposition 1. By Lemma 1, CARA and CRRA utilities satisfy this condition, and thus Proposition 1 follows immediately from these lemmas.

Let $S >_c R$ satisfying (6) for some $M \in \mathbb{R}$. The key step of the proof of Lemma 2 is to construct $S' >_c R'$ such that

$$\frac{U_B(S) - U_B(R)}{U_A(S) - U_A(R)} = \frac{U_B(S') - U_B(R')}{U_A(S') - U_A(R')},$$

while making the utility differences arbitrarily small. To this end, we consider the menu $(S, R_\lambda S)$, where $\lambda \in (0, 1)$ and $R_\lambda S$ is distributed as a compound lottery that yields R with probability λ and S with probability $1 - \lambda$. As $\lambda \rightarrow 0$, the utility difference between S and $R_\lambda S$ vanishes. Lemma 2 exploits the linearity of vNM utilities, which ensures that the ratio of utility differences is independent of λ . It also relies on the differentiability of F_A and F_B to approximate them around 0 by affine functions. Lemma 2 is proved in Appendix A.

Note that by Lemma 1, (6) is satisfied for all concave-ordered lotteries under sufficient shifting/scaling. Moreover, for CARA, the utility difference $U(S + x) - U(R + x)$ tends to 0 as x tends to infinity (see Lemma 4 in Appendix B). For CRRA, on the other hand, $U(kS) - U(kR)$ increases with k for relative risk coefficients less than unity. We therefore apply an additional transformation of scaling down the

probability of receiving the risky prospect so that the utility difference is arbitrarily small. Based on these observations, the following proposition extends the non-monotonicity result of Proposition 2 to all $F_A, F_B \in \mathcal{F}$.

Proposition 3. *Let $F_A, F_B \in \mathcal{F}$ and $S >_c R$. Let U_A and U_B be distinct CARA utilities and V_A and V_B be distinct CRRA utilities. Then there exist $x_0, k_0 > 0$ such that*

1. $S + x$ and $R + x$ generate a reversal for (U_A, F_A) and (U_B, F_B) for $x > x_0$.
2. $k \cdot S$ and $k \cdot (R_\lambda S)$ generate a reversal for (V_A, F_A) and (V_B, F_B) for $k > k_0$ and λ small enough.

Proposition 3 is proved in Appendix B. While CARA preferences are invariant to changes in background wealth, the above proposition shows that non-monotonicity occurs in FEU models under CARA utilities for any concave-ordered lotteries under sufficient background wealth. Likewise, while CRRA preferences are invariant to the scaling of stakes and to mixing with the safe lottery, these transformations can always generate a non-monotonicity.

Proposition 3 demonstrates that non-monotonicity is a pervasive problem. It holds for all pairs of risk parameters, for all pairs of Fechnerian noise specifications, and for all concave-ordered lotteries after a transformation that preserves the concave ordering as well as the family of preferences being measured.

4 Weak Utility Models

In the previous section, we studied models in which noise varies across individuals but not across menus for the same individual. In this section, we study a more flexible model that incorporates menu-dependent noise and show that the reversals under CARA/CRRA utilities persist.

Formally, ρ has a *weak expected-utility* (WEU) representation (U, F, σ) if

$$\rho(X, Y) = F\left(\frac{U(X) - U(Y)}{\sigma(X, Y)}\right), \quad (7)$$

where U is a vNM utility function, $F \in \mathcal{F}$ is a Fechnerian noise structure, and $\sigma: \mathcal{L} \times \mathcal{L} \rightarrow (0, \infty)$ is a menu-dependent scale parameter satisfying $\sigma(X, Y) = \sigma(Y, X)$. Symmetry of σ is required so that $\rho(X, Y) + \rho(Y, X) = 1$.

An example of WEU is the *menu-dependent probit model*. In this model, the shock terms ε and ε' in (2) follow normal distributions, but the identity assumption is relaxed to allow their variances to depend on the lotteries. That is,

$$\rho(X, Y) = \mathbb{P}(U(X) + \varepsilon_X \geq U(Y) + \varepsilon_Y),$$

where ε_X and ε_Y are independent and normal random variables with means zero and variances σ_X and σ_Y .¹⁵ For example, σ_X may depend on the variance of lottery X in the menu. This can capture the intuitive idea that more variable lotteries may generate more noise, allowing for mistake probabilities to remain high even when stakes are high.

Another example is the *strict utility model* of Luce and Suppes (1965) in which the choice probabilities are determined by the utility ratio

$$\rho(X, Y) = \frac{U(X)}{U(X) + U(Y)},$$

rather than by the utility difference as in FEU models (see (1)) of the previous section. Indeed, any *simply scalable expected utility* (SSEU) representation, which takes the form

$$\rho(X, Y) = H(U(X), U(Y)), \tag{8}$$

where $H : \mathbb{R}^2 \rightarrow [0, 1]$ satisfies $H(s, t) + H(t, s) = 1$ and is strictly increasing in its first argument, has a WEU representation.¹⁶

As the examples above suggest, the menu-dependent scale parameter σ allows for a highly flexible stochastic choice model. In fact, without imposing restrictions on σ , the sole behavioral implication of WEU is that individuals select their preferred lottery with probability greater than one-half.¹⁷ In order to rule out pathological noise

¹⁵In this case, ρ can be written as (7) where F is the CDF of the standard normal distribution and $\sigma(X, Y) = \sqrt{\sigma_X^2 + \sigma_Y^2}$.

¹⁶To see that the simply scalable model is a special case of the weak utility model, take

$$\sigma(X, Y) = \begin{cases} \frac{U(X) - U(Y)}{F^{-1}(H(U(X), U(Y)))} & U(X) \neq U(Y) \\ s > 0 & U(X) = U(Y) \end{cases}$$

in (7) to derive $\rho(X, Y) = H(U(X), U(Y))$.

¹⁷Since $F(t/\sigma(X, Y)) \in \mathcal{F}$ for all $(X, Y) \in \mathcal{L} \times \mathcal{L}$, if $U(X) - U(Y) \geq 0$, then $F\left(\frac{U(X) - U(Y)}{\sigma(X, Y)}\right) \geq \frac{1}{2}$ with equality if and only if $U(X) = U(Y)$. Conversely, if $U(X) > U(Y)$, then for any $p \in (\frac{1}{2}, 1)$, letting $\sigma(X, Y) = \frac{U(X) - U(Y)}{F^{-1}(p)}$ yields the desired probability.

assignments, we require that menus with similar lotteries are associated with similar menu-dependent scale parameters. We say that a sequence of lotteries $X_1, X_2, \dots$ converges to a lottery X if for all continuous functions u it holds that

$$\mathbb{E}[|u(X_n) - u(X)|] \rightarrow 0.$$

In other words, $X_1, X_2, \dots$ converges to X if the sequence of random utilities $u(X_1), u(X_2), \dots$ converges to $u(X)$ in L^1 for every continuous Bernoulli u . We say that σ is *continuous* if whenever $X_n \rightarrow X$ and $Y_n \rightarrow Y$, it holds that $\sigma(X_n, Y_n) \rightarrow \sigma(X, Y)$. We say a WEU representation (U, F, σ) is *continuous* when σ is continuous.¹⁸

Importantly, vNM utilities are continuous, i.e., $U(X_n) \rightarrow U(X)$ whenever $X_n \rightarrow X$, so we can allow σ to depend continuously on U . Moreover, σ is not required to depend only on the marginal distributions of the lotteries in the menu; it may, for example, depend on their joint distribution.

4.1 Characterization

Our next theorem characterizes when two vNM utilities necessarily lead to reversals under any continuous WEU models. Given two increasing and concave Bernoulli utility functions u_A and u_B , we say that u_A is *absolutely more concave than* u_B , and write $u_A \geq_{\text{abs}} u_B$ if $u_A - u_B$ is concave. Moreover, we say that u_A is *relatively more concave than* u_B , and write $u_A \geq_{\text{rel}} u_B$ if there exists $k > 0$ such that $k \cdot u_A(x) - u_B(x)$ is concave.¹⁹ We extend these notions to vNM utilities $U_A = \mathbb{E}[u_A]$ and $U_B = \mathbb{E}[u_B]$, writing $U_A \geq_{\text{abs}} U_B$ if $u_A \geq_{\text{abs}} u_B$ and $U_A \geq_{\text{rel}} U_B$ if $u_A \geq_{\text{rel}} u_B$. For twice differentiable functions with $u_A'' < 0$, the condition that $u_A \geq_{\text{rel}} u_B$ is equivalent to the boundedness of $u_B''(x)/u_A''(x)$, i.e., that $u_B'' = O(u_A'')$ on $\mathbb{R}$.

¹⁸Formally, we require σ to be sequentially continuous in the product topology on $\mathcal{L} \times \mathcal{L}$, when $\mathcal{L}$ is equipped with the coarsest topology making each $X \mapsto u(X)$ continuous as an L^1 -valued map. This topology is very fine (finer than any L^p topology for $1 \leq p < \infty$), making many functions on $\mathcal{L}$ continuous.

¹⁹Recall that u_A is more risk-averse than u_B in the Arrow-Pratt sense if $u_A = f \circ u_B$ for some increasing and concave function f . This notion of comparative concavity is logically independent of both $\geq_{\text{abs}}$ and $\geq_{\text{rel}}$. In Appendix D, we show that the stronger comparative risk notion of Ross (1981) implies $\geq_{\text{rel}}$ and discuss an interpretation of his stronger notion in the stochastic framework.

Theorem 1. *Let U_A and U_B be vNM utilities with U_A more risk-averse and let $F_A, F_B \in \mathcal{F}$. There exist continuous menu-dependent scale parameters σ_A and σ_B for which no reversals occur if and only if $U_A \geq_{\text{rel}} U_B$.*

The necessity of $U_A \geq_{\text{rel}} U_B$ to avoid reversals is proved in Appendix C. A key idea is to show that if U_A is not relatively more concave than U_B , then the ratio

$$\frac{U_B(S) - U_B(R)}{U_A(S) - U_A(R)}$$

is unbounded over the set of lottery pairs $S >_c R$. Under continuity of σ_A and σ_B , we show that this unboundedness leads to reversals using an argument similar to that of Proposition 1. This argument relies on the fact that continuity imposes a bound on the relative noisiness of A and B for some lottery comparisons. We note that limitations on relative noisiness lead to reversals even under some discontinuous menu-dependent scale parameters. For example, for (U_A, F, σ_A) and (U_B, F, σ_B) , if $\sigma_B(S, R)/\sigma_A(S, R)$ is bounded, there will necessarily be reversals even if σ_A and σ_B are not continuous.

We prove sufficiency of $U_A \geq_{\text{rel}} U_B$ here by showing that even under a simple heteroskedastic probit model, without menu dependence, this condition suffices. Indeed, suppose that $ku_A - u_B$ is concave for some $k > 0$. Let F be the CDF of the standard normal distribution. Let $\sigma_A(X, Y) = 1/k$ and $\sigma_B(X, Y) = 1$ for all X, Y . Then for any lotteries $S \geq_c R$,

$$\rho_A(S, R) = F(k(U_A(S) - U_A(R))) \quad (9)$$

$$\rho_B(S, R) = F(U_B(S) - U_B(R)). \quad (10)$$

Since $ku_A - u_B$ is concave, $kU_A(S) - U_B(S) \geq kU_A(R) - U_B(R)$. Hence, since F is strictly increasing, (9) exceeds (10), meaning that there are no reversals.²⁰

Theorem 1 establishes a simple test for identifying whether a pair of vNM utilities generates reversals. Indeed, for CARA and CRRA utilities, $\text{cara}_b''(x)/\text{cara}_a''(x) = (b/a)e^{-(b-a)x}$ and $\text{crra}_b''(x)/\text{crra}_a''(x) = (b/a)x^{a-b}$ are unbounded for $a > b$. Thus, it follows from Theorem 1 that reversals of Fechnerian CARA and CRRA models cannot be eliminated by allowing continuous menu-dependent scale parameters.

Corollary 1. *Let $F_A, F_B \in \mathcal{F}$ and let σ_A and σ_B be continuous. Then for any distinct CARA (CRRA) utilities U_A and U_B there are reversals.*

²⁰Note that for this direction, we can relax the requirement that $F \in C_+^1$.

Corollary 1 highlights that the non-monotonicity result of the menu-independent FEU models (Proposition 1) is not merely a byproduct of asymptotic properties of choice probabilities when stakes are increased. Indeed, in the following example scale parameters are engineered so that choice probabilities are scale-invariant, yet monotonicity does not obtain.

Example 1. Let $a > b > 0$, $U_A = \text{CRRA}_a$, and $U_B = \text{CRRA}_b$. Let F be the CDF of the standard normal distribution. Let

$$\sigma_A(X, Y) = \mathbb{E}[X^{a-1} + Y^{a-1}]^{-1}, \quad \sigma_B(X, Y) = \mathbb{E}[X^{b-1} + Y^{b-1}]^{-1}.$$

Then for all $k > 0$ and for $i \in \{A, B\}$, we have

$$F\left(\frac{U_i(kX) - U_i(kY)}{\sigma_i(kX, kY)}\right) = F\left(\frac{U_i(X) - U_i(Y)}{\sigma_i(X, Y)}\right).$$

When $a \geq 1$, σ_A decreases as X and Y are scaled up, capturing the behavioral property that individuals pay more attention when facing higher stakes. Moreover, the standard deviations are chosen so that each individual's probability of choosing βX over βY does not depend on $\beta > 0$, as $\sigma_A(\beta X, \beta Y) = \beta^{1-a} \sigma_A(X, Y)$ and

$$U_A(\beta X) - U_A(\beta Y) = \beta^{1-a} (U_A(X) - U_A(Y)).$$

Since $\sigma_A(X_n, Y_n) \rightarrow \sigma_A(X, Y)$ whenever $X_n \rightarrow X$ and $Y_n \rightarrow Y$, the model is continuous. Thus, it follows from Corollary 1 that even these calibrated models cannot avoid the paradox.

Another example of continuous WEU is when ρ admits an SSEU representation (U, H) given in (8), above, where H is further assumed to be differentiable and its first partial derivative, H_1 , is continuous and positive. Indeed, for

$$\sigma(X, Y) = \begin{cases} \frac{U(X) - U(Y)}{F^{-1}(H(U(X), U(Y)))} & U(X) \neq U(Y) \\ \frac{F'(F^{-1}(H(U(Y), U(Y))))}{H_1(U(Y), U(Y))} & U(X) = U(Y), \end{cases}$$

the WEU model (U, F, σ) coincides with the SSEU model (U, H) . Moreover, σ is continuous since U , F^{-1} , F' , H , and H_1 are continuous and the two cases coincide in the limit as $U(X_n)$ and $U(Y_n)$ tend to the same value.

One may guess, at this point, that reversals arise in WEU models under CARA and CRRA utilities because the Fechnerian noise and menu-dependent scale parameter are applied to the vNM utility U , which is merely a representation of the preference and carries no cardinal significance. A natural alternative would be to calculate, for each lottery X , its certainty equivalent $CE(X) = u^{-1}(\mathbb{E}[u(X)])$, and let

$$\rho(X, Y) = F \left(\frac{CE(X) - CE(Y)}{\sigma(X, Y)} \right).$$

However, the next result demonstrates that this adjustment does not resolve the issue. In fact, reversals persist even if we more generally consider utility representations of the form $f \circ U$ for an arbitrary $f \in C_+^1$.

Corollary 2. *Let $F_A, F_B \in \mathcal{F}$, $f_A, f_B \in C_+^1$ and let σ_A and σ_B be continuous. For any distinct CARA (CRRA) utilities U_A and U_B , there are reversals for $(f_A \circ U_A, F_A, \sigma_A)$ and $(f_B \circ U_B, F_B, \sigma_B)$.*

Corollary 2 shows that reversals are a fundamental feature of CARA and CRRA preferences, not an artifact of the utility representation. The underlying mechanism is straightforward: for any transformation $f \in C_+^1$, the WEU model $(f \circ U, F, \sigma)$ has a WEU representation (U, F, τ) , where the new menu-dependent scale parameter τ remains continuous.²¹ Given this, the result follows directly from Corollary 1. Since inverses of cara and crra belong to C_+^1 , replacing CARA and CRRA expected utilities with their certainty equivalents cannot eliminate reversals.

Corollaries 1 and 2 demonstrate that CARA and CRRA utilities generate reversals whenever choices are noisy, even if we use very general utility representations and allow noise structures to vary across individuals and menus. Thus, if we hope to model the stochastic choice behavior of A and B , where A consistently chooses safer lotteries more frequently than B , we must forgo CARA and CRRA preferences. Moreover, the boundedness of u_B''/u_A'' is violated by many parametric families of utilities that are used to model risk aversion. In particular, we show in Appendix E

²¹Here,

$$\tau(X, Y) = \begin{cases} \frac{U(X) - U(Y)}{f(U(X)) - f(U(Y))} \sigma(X, Y) & U(X) \neq U(Y) \\ \frac{\sigma(X, Y)}{f'(U(Y))} & U(X) = U(Y), \end{cases}$$

which is continuous since U , f , f' , and σ are continuous and the two cases coincide in the limit.

that expo-power utility—a popular generalization of the CARA/CRRA families—violates the condition, leading to reversals under any continuous WEU models. We take on these challenges in the next section, where we suggest alternative vNM utilities that can accommodate consistent comparative statics under standard models of noise.

5 Monotone Alternatives to CARA and CRRA

In the previous section, we showed that when two vNM utilities have an unbounded ratio of second derivatives, there are necessarily reversals under very general noise structures. Conversely, when the ratio of second derivatives is bounded, there exist heteroskedastic noise structures such that there are no reversals. In this section, we demonstrate when two vNM utilities will not lead to reversals under any homoskedastic Fechnerian models and, more generally, under homoskedastic WEU models. We provide two parametric families of utilities such that any two utilities within either family satisfy this condition. We then discuss the economic properties of these utility families.

Recall that $u_A \geq_{\text{abs}} u_B$ if $u_A - u_B$ is concave. The next proposition states that this absolute comparative concavity notion leads to no reversals under identical Fechnerian noise.

Proposition 4. *Let $F \in \mathcal{F}$ and fix U_A more risk-averse than U_B . Then there are no reversals under (U_A, F) and (U_B, F) if and only if $U_A \geq_{\text{abs}} U_B$.*

Proof of Proposition 4. Since Fechnerian noise structures $F \in \mathcal{F}$ are strictly increasing, the choice probability of S is higher for A than B if and only if the utility difference is higher for A than B , i.e.,

$$U_A(S) - U_A(R) \geq U_B(S) - U_B(R),$$

or equivalently,

$$U_A(S) - U_B(S) \geq U_A(R) - U_B(R). \tag{11}$$

We now show that this inequality holds for all lotteries $S \geq_c R$ if and only if $U_A \geq_{\text{abs}} U_B$.

Let $h = u_A - u_B$. If h is concave, then (11) is satisfied by the definition of the concave order. Conversely, for the sake of contradiction, suppose that h is not concave. Then there are $x, y \in \mathbb{R}$ and $\lambda \in (0, 1)$ such that

$$h(\lambda x + (1 - \lambda)y) < \lambda h(x) + (1 - \lambda)h(y).$$

Let S denote a degenerate lottery that always pays $\lambda x + (1 - \lambda)y$, and let R denote a lottery that pays x with probability λ and y with complementary probability. Then

$$U_A(S) - U_B(S) = h(\lambda x + (1 - \lambda)y) < \lambda h(x) + (1 - \lambda)h(y) = U_A(R) - U_B(R),$$

contradicting (11). Thus $u_A - u_B$ is concave, meaning $U_A \geq_{\text{abs}} U_B$. $\square$

In the case where A and B have different noise structures, i.e., $F_A \neq F_B$, a sufficient condition for no reversals is if $U_A \geq_{\text{abs}} U_B$ and F_A is *more precise than* F_B , in the sense that $F_A(t) \geq F_B(t)$ for $t \geq 0$. A more precise noise structure translates the same utility difference into a higher probability of choosing a more preferred alternative. Since A is risk-averse, increasing A 's precision will only increase the probability that A chooses S . In particular,

$$F_A(U_A(S) - U_A(R)) \geq F_A(U_B(S) - U_B(R)) \geq F_B(U_B(S) - U_B(R)),$$

where the first inequality follows from Proposition 4.²² Of course, A may experience much more noise than B , in which case A will make more mistakes and may choose riskier options more frequently than B , even with $U_A \geq_{\text{abs}} U_B$.

5.1 Parametric Families of Utility Functions

We conclude by proposing parametric utility families ordered by both risk aversion and $\geq_{\text{abs}}$, leading to monotone comparative statics under identical (menu-dependent) Fechnerian noise. Recall that for strictly increasing concave functions f and g , f is *more concave* than g if $f \circ g^{-1}$ is increasing and concave. The following proposition provides a simple way to construct a parametric utility family that yields consistent comparative statics.

²²The same conclusion holds under WEU models when A and B face the same menu-dependent scale parameters or when $\sigma_A(S, R) \leq \sigma_B(S, R)$ for all $S \geq_c R$.

Proposition 5. *Let f and g be strictly increasing concave functions such that f is more concave than g . Let $u_A(x) = af(x) + g(x)$ and $u_B(x) = bf(x) + g(x)$, where $a > b \geq 0$. Then $U_A = \mathbb{E}[u_A]$ is more risk-averse than $U_B = \mathbb{E}[u_B]$ and $U_A \geq_{\text{abs}} U_B$.*

These utilities are parametrized by the coefficient on the more risk-averse Bernoulli utility function f , i.e., the higher the coefficient, the greater the risk aversion.²³ Moreover, $u_A - u_B = (a - b)f$ is concave since f is concave, so $u_A \geq_{\text{abs}} u_B$, and increasing, since f is increasing and $a > b$. Thus, monotonicity of choice probabilities in the risk coefficient also holds for all lotteries comparable by second-order stochastic dominance (SOSD). We write $X \geq_{\text{ssd}} Y$ if X dominates Y in the SOSD order.

Remark 2. *Let f and g be strictly increasing concave functions such that f is more concave than g . Let $u_A(x) = af(x) + g(x)$ and $u_B(x) = bf(x) + g(x)$, where $a > b \geq 0$. Then for any $F \in \mathcal{F}$,*

$$F(U_A(X) - U_A(Y)) \geq F(U_B(X) - U_B(Y)) \quad (12)$$

for all $X \geq_{\text{ssd}} Y$.

The contrast between the utility families in Proposition 5 and standard functional forms, such as CARA and CRRA, is stark. Under Fechnerian noise, the latter admit reversals under the concave order, whereas our proposed families preclude reversals even under the more general criterion of second-order stochastic dominance.

When f and g are CARA (CRRA) utilities with different coefficients, u_A and u_B correspond to sum-ex (sum-power) utilities (see, e.g., Farquhar and Nakamura, 1987; Bell, 1988; Pedersen and Satchell, 2001). Proposition 5 thus suggests a replacement of the CARA utility family or CRRA utility family for estimation. One specifies an upper and lower bound on absolute risk aversion, corresponding to two CARA utilities, and then estimates the weight on each utility. The same exercise can be done with bounds on relative risk aversion and CRRA utilities. In Appendix F, we discuss properties of sum-ex and sum-power utility functions and how to interpret the parameters.

²³Indeed,

$$-\frac{u_A''(x)}{u_A'(x)} \geq -\frac{u_B''(x)}{u_B'(x)} \iff (a - b) \left(-\frac{f''(x)}{f'(x)} + \frac{g''(x)}{g'(x)} \right) \geq 0.$$

Since f is more concave than g , it follows that $a \geq b$ is equivalent to U_A being more risk-averse than U_B .

6 Multinomial Choice

In this section, we extend our results to multinomial choice. Given a finite menu $M \subset \mathcal{L}$, we denote by $\rho^M \in \Delta(M)$ the choice probabilities for the menu M .

There are a number of ways to extend the notions of a reversal to multinomial choice. We discuss two notions that seem natural.

Definition 2. *When A is more risk-averse than B , we say that there is a compound reversal if there is a menu M such that for all concave functions $u: \mathbb{R} \rightarrow \mathbb{R}$*

$$\sum_{X \in M} \mathbb{E}[u(X)]\rho_B^M(X) \geq \sum_{X \in M} \mathbb{E}[u(X)]\rho_A^M(X),$$

with strict inequality for at least one such u .

When a compound reversal occurs at M , the compound lottery resulting from A 's stochastic choice is dominated by that of B in the concave order. The following alternative notion requires that for each set $N \subset M$ that is upward-closed in the concave order,²⁴ B is at least as likely as A to choose a lottery in N and strictly more likely for at least one such N .

Definition 3. *When A is more risk-averse than B , we say that there is a FOSD reversal if there is a menu M such that for all functions $f: \mathcal{L} \rightarrow \mathbb{R}$ that are increasing in $\geq_c$*

$$\sum_{X \in M} f(X)\rho_B^M(X) \geq \sum_{X \in M} f(X)\rho_A^M(X),$$

with strict inequality for at least one such f .

By choosing functions f that are expectations of strictly concave real functions in Definition 3, we see that an FOSD reversal is always a compound reversal. The converse holds only for menus of size two, in which case both notions coincide with Definition 1.

6.1 Additive Perturbed Utility Models

In this section, we study multinomial choice reversals in the *additive perturbed utility* (APU) framework of Fudenberg et al. (2015), wherein

$$\rho^M = \operatorname{argmax}_{p \in \Delta(M)} \sum_{X \in M} U(X)p(X) - c(p(X)), \quad (13)$$

²⁴ N is upward closed in the concave order if for all $S \geq_c R$, $R \in N$ implies $S \in N$.

where U is a vNM utility function and $c : [0, 1] \rightarrow \mathbb{R} \cup \{\infty\}$ is strictly convex and continuously differentiable on $(0, 1)$ and satisfies $\lim_{p \rightarrow 0} c'(p) = -\infty$. Any function c with these properties is referred to as a *cost function*. Both U and c are fixed across menus. We say that ρ has an *APU representation* (U, c) if ρ coincides with (13). Fudenberg et al. (2015) show that APU is a generalization of Fechnerian choice to the multinomial setting; see also Appendix G.2.

Our next result shows that reversals occur in APU models with arbitrarily heterogeneous cost functions unless the $\geq_{\text{rel}}$ order obtains.

Theorem 2. *Let U_A and U_B be vNM utilities with U_A more risk-averse. The following are equivalent:*

1. *There exist cost functions $c_A, c_B \in C_+^2$ for which no compound reversals occur;*
2. *There exist cost functions $c_A, c_B \in C_+^2$ for which no FOSD reversals occur;*
3. $U_A \geq_{\text{rel}} U_B$.

Theorem 2 is proved in Appendix G, in which we further show that when ρ_A and ρ_B have APU representations with the same cost function, reversals of both kinds will occur in some menus unless $U_A \geq_{\text{abs}} U_B$, extending Proposition 4 to the multinomial setting (Proposition 10). Given that APU is an extension of the FEU (see Lemma 7), we already know from the earlier sections that reversals will occur under heterogeneous (homogeneous) costs if the $\geq_{\text{rel}}$ ($\geq_{\text{abs}}$) order does not obtain. The message of this section is thus that when these orders do obtain, reversals can be avoided even for multinomial choice.

6.2 Random Parameter Models

As discussed above, Apesteguia and Ballester (2018) raise the issue of reversals for CARA and CRRA models of binary choice under homogeneous Fechnerian noise. As an alternative, they advocate for *random parameter* models, in which each individual's vNM utility is randomly drawn from a parameterized family. Formally, an *ordered vNM family* is an indexed set of risk-averse vNM utilities $\mathcal{U} = \{U_a\}_{a \in \mathcal{A}}$ for some $\mathcal{A} \subset \mathbb{R}$ and is ordered by risk aversion, i.e., for $a > b$, U_a is more risk-averse than U_b . In a random parameter model on $\mathcal{U}$, each individual i is associated with a distribution $\mu_i \in \Delta(\mathcal{A})$. We denote by ξ_i a random draw from μ_i with corresponding

random vNM utility U_{ξ_i} . In this model, individual i chooses lottery X over Y with probability²⁵

$$\rho_i(X, Y) = \mathbb{P}(U_{\xi_i}(X) \geq U_{\xi_i}(Y)).$$

We say that A is more risk-averse than B if μ_A dominates μ_B in the sense of first-order stochastic dominance, denoted $\mu_A >_{\text{FOSD}} \mu_B$.²⁶ Clearly if $X >_c Y$, then every realization of U_{ξ_i} would evaluate X greater than Y , since every possible realized utility function exhibits risk aversion. Since lotteries dominated in the concave order are never chosen in these models, there cannot, of course, be reversals.²⁷

Random parameter models are often studied in multinomial settings, where

$$\rho_i^M(X) = \mathbb{P}(U_{\xi_i}(X) = \max_{Y \in M} U_{\xi_i}(Y)).$$

While these models preclude reversals in binary menus, in which case compound reversals and FOSD reversals coincide, the next results show that the situation is more subtle for larger menus.

Proposition 6. *For any ordered vNM family $\mathcal{U}$ and any random parameter models on $\mathcal{U}$, there are no FOSD reversals.*

The reason that random parameter models do not lead to FOSD reversals is much the same as the reason that they do not admit reversals in binary menus. To see this formally, suppose that individual A has random utility U_{ξ_A} and B has random utility U_{ξ_B} , where A is more risk-averse than B . Fix a menu M and let $N \subset M$ denote the set of lotteries that A chooses with positive probability. Since A never chooses dominated lotteries, for $Y \in N$, the indicator function $\mathbb{1}_Y$ is increasing in $\geq_c$. Thus, for there to be an FOSD reversal,

$$\rho_B^M(Y) = \sum_{X \in M} \mathbb{1}_Y \rho_B^M(X) \geq \sum_{X \in M} \mathbb{1}_Y \rho_A^M(X) = \rho_A^M(Y).$$

²⁵Technically, this is not a well-defined probabilistic choice rule when the probability of ties is non-zero. Rather than define a tie-breaking rule, we restrict ourselves to comparisons (and later on menus) in which ties occur with zero probability.

²⁶This FOSD requirement is the natural ordinal generalization of the additive shift model $(a + \varepsilon)$ used by Apesteguia and Ballester (2018). The additive shift model $a + \varepsilon$ implicitly relies on the cardinal scale of the parameter space. The natural ordinal generalization is to compare individuals' distributions by FOSD.

²⁷Indeed, Apesteguia and Ballester (2018) point out that random parameter models have limitations when applied to stochastic-dominance related lotteries. They suggest adding an additional trembling stage as a way to generate stochasticity for such pairs.

Since this must hold for all $Y \in N$, we conclude that $\rho_B^M = \rho_A^M$, which precludes an FOSD reversal since the defining inequality holds with equality for all increasing f .

Compound reversals, however, do arise for some ordered vNM families. Importantly, CARA- and CRRA-based random parameter models suffer from compound reversals (see Appendix H.1 for examples). The key step is to construct a menu $\{X, Y, Z\}$, where Y dominates $X_{\frac{1}{2}}Z$ in the concave order, and the most preferred alternative shifts from Y to X to Z as the risk coefficient increases. Let A have random parameters supported on the lower region of risk coefficients, where Y is the most preferred alternative. Moreover, let the random parameters of B be supported on the higher range of coefficients, where the most preferred alternatives are X and Z , assigning equal and strictly positive probability mass to the regions where each is optimal. The resulting compound lottery from A 's choice distribution then dominates that of B in the concave order, yielding a compound reversal.

Interestingly, random parameter models based on sum-ex and sum-power utilities—and more generally, utilities of the form in Proposition 5—do not yield compound reversals. In fact, our next result shows that utilities of the form in Proposition 5 are essentially the only utility families that do not admit compound reversals.

We say that an ordered vNM family $\mathcal{U} = \{U_a\}_{a \in \mathcal{A}}$ is *collinear* if for each U_a there is a positive affine transformation $U_a \mapsto V_a$ such that the transformed family $\{V_a\}_{a \in \mathcal{A}}$ lies on a line. I.e., for any $a > b > c$, there is $\alpha \in (0, 1)$ such that $V_b = \alpha V_a + (1 - \alpha)V_c$.

Theorem 3. *Let $\mathcal{U} = \{U_a\}_{a \in \mathcal{A}}$ be an ordered vNM family. The following are equivalent:*

1. *For any $\mu_A, \mu_B \in \Delta(\mathcal{A})$ and for any menu M , there are no compound reversals;*
2. *$\mathcal{U}$ is collinear.*

Theorem 3 is proved in Appendix H.2. We note that the first statement can be replaced by a quantification over continuous μ_A and μ_B , provided that $U_a(X)$ is continuous in the parameter for all X . This condition holds for many standard parametric classes, including the CARA and CRRA families.

7 Conclusion

This paper establishes that the paradox long noted for CARA and CRRA utilities under homoskedastic Fechnerian noise is not an artifact of restrictive parametric

assumptions, but instead reflects a deeper incompatibility between these utility forms and noisy choice. We show that the same non-monotone comparative statics arise under highly flexible noise specifications—including heterogeneous noise across individuals and menu-dependent noise.

We obtain a simple characterization of whether two utility functions can yield sensible predictions in the presence of noise. This condition rules out even the more general expo-power utilities, and points to new parametric families that restore intuitive comparative statics. These families, which include sum-ex and sum-power utilities, provide empirically tractable, well-behaved alternatives for measuring risk preferences in noisy environments. We leave to future work an empirical examination of how risk aversion and decision noise co-vary across different stake levels, as well as systematic tests of these proposed utility families under a range of noise specifications.

We conclude by noting that the continuity requirement on noise assignments may be restrictive in some settings. For example, discontinuous noise assignments may capture interesting features of binary comparisons, such as salience or complexity. Understanding which discontinuous models generate reversals thus remains an interesting direction for future research.

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A Proof of Proposition 1

We start by proving Lemma 1, which shows that

$$\frac{U_B(S) - U_B(R)}{U_A(S) - U_A(R)}$$

is unbounded over the set of pairs $S >_c R$.

Proof of Lemma 1. Let $S >_c R$ and $a, b > 0$. For CARA, we have

$$\begin{aligned} \frac{\text{CARA}_b(S+x) - \text{CARA}_b(R+x)}{\text{CARA}_a(S+x) - \text{CARA}_a(R+x)} &= \frac{a \mathbb{E}[e^{-b(R+x)}] - \mathbb{E}[e^{-b(S+x)}]}{b \mathbb{E}[e^{-a(R+x)}] - \mathbb{E}[e^{-a(S+x)}]} \\ &= \frac{a}{b} \left(\frac{\mathbb{E}[e^{-bR}] - \mathbb{E}[e^{-bS}]}{\mathbb{E}[e^{-aR}] - \mathbb{E}[e^{-aS}]} \right) e^{(a-b)x}. \end{aligned}$$

For CRRA, when a and b are not equal to 1, we have

$$\begin{aligned} \frac{\text{CRRA}_b(k \cdot S) - \text{CRRA}_b(k \cdot R)}{\text{CRRA}_a(k \cdot S) - \text{CRRA}_a(k \cdot R)} &= \frac{1-a}{1-b} \left(\frac{\mathbb{E}[S^{1-b}] - \mathbb{E}[R^{1-b}]}{\mathbb{E}[S^{1-a}] - \mathbb{E}[R^{1-a}]} \right) \frac{k^{1-b}}{k^{1-a}} \\ &= \frac{1-a}{1-b} \left(\frac{\mathbb{E}[S^{1-b}] - \mathbb{E}[R^{1-b}]}{\mathbb{E}[S^{1-a}] - \mathbb{E}[R^{1-a}]} \right) k^{a-b}. \end{aligned}$$

Since positive risk coefficients a and b for CARA/CRRA utilities correspond to risk-averse utilities,

$$C_1 = \frac{a}{b} \left(\frac{\mathbb{E}[e^{-bR}] - \mathbb{E}[e^{-bS}]}{\mathbb{E}[e^{-aR}] - \mathbb{E}[e^{-aS}]} \right), \quad C_2 = \frac{1-a}{1-b} \left(\frac{\mathbb{E}[S^{1-b}] - \mathbb{E}[R^{1-b}]}{\mathbb{E}[S^{1-a}] - \mathbb{E}[R^{1-a}]} \right) \quad (14)$$

are positive. For a or b equal to 1, the expression for C_2 is given by the corresponding limit as a or b tends to 1. $\square$

Lemma 1 provides sequences of lottery pairs (S_n, R_n) along which the ratio of expected-utility differences $(U_B(S_n) - U_B(R_n))/(U_A(S_n) - U_A(R_n))$ diverges, for CARA and CRRA utilities. Lemma 2 establishes that when such a sequence exists there will be reversals. An important subtlety of Lemma 2 is that the reversal-generating lotteries may not be part of this sequence. Indeed, Lemma 2 relies on the following lemma, which establishes that under the additional condition that utility differences tend to zero, there will be lotteries in the sequence that generate reversals.

Lemma 3. *Let $F_A, F_B \in \mathcal{F}$ and let U_A and U_B be risk-averse vNM utility functions with U_A more risk-averse. There exist $M \in \mathbb{R}$ and $\delta > 0$ such that all lotteries $S >_c R$ satisfying*

$$\frac{U_B(S) - U_B(R)}{U_A(S) - U_A(R)} \geq M \quad \text{and} \quad U_i(S) - U_i(R) < \delta$$

for $i = A, B$ generate a reversal.

Proof of Lemma 3. Let $f_A(t) = \frac{d}{dt}F_A(t)$ and $f_B(t) = \frac{d}{dt}F_B(t)$. Since F_A is continuously differentiable, f_A is continuous at 0 so that for each $\varepsilon_A > f_A(0)$ there is $\delta_A > 0$ such that for each $t \in (0, \delta_A)$, $f_A(t) < \varepsilon_A$ and $F_A(t) < \frac{1}{2} + t\varepsilon_A$. Since F_B is continuously differentiable and $f_B > 0$, for each $0 < \varepsilon_B < f_B(0)$, there exists $\delta_B > 0$ such that for any $t \in (0, \delta_B)$, $f_B(t) > \varepsilon_B$ and $F_B(t) > \frac{1}{2} + t\varepsilon_B$. Let $\delta = \min\{\delta_A, \delta_B\}$ and let $M = \varepsilon_A/\varepsilon_B$.

Let S and R as in the statement of the lemma. Then

$$\begin{aligned} F_B(U_B(S) - U_B(R)) &> \frac{1}{2} + (U_B(S) - U_B(R))\varepsilon_B \\ &\geq \frac{1}{2} + (U_A(S) - U_A(R))\varepsilon_A \\ &> F_A(U_A(S) - U_A(R)). \end{aligned}$$

□

In light of Lemma 3, Lemma 2 follows from the observation that for any lotteries $S >_c R$ and $\varepsilon > 0$, there exist lotteries $S' >_c R'$ with the same utility difference ratio such that the each utility difference is less than ε .

Proof of Lemma 2. Let $F_A, F_B \in \mathcal{F}$ and let U_A and U_B be risk-averse vNM utility functions with U_A more risk-averse. By Lemma 3, there exist $M \in \mathbb{R}$ and $\delta > 0$ such that all lotteries $S >_c R$ satisfying

$$\frac{U_B(S) - U_B(R)}{U_A(S) - U_A(R)} \geq M \quad \text{and} \quad U_i(S) - U_i(R) < \delta$$

for $i = A, B$ generate a reversal. By hypothesis, there exist lotteries $S >_c R$ satisfying the first inequality.

For $\lambda \in (0, 1)$, let $R_\lambda S$ denote a lottery distributed as a compound lottery that yields R with probability λ and yields S with probability $1 - \lambda$.

Then $S >_c R_\lambda S$ and

$$\frac{U_B(S) - U_B(R_\lambda S)}{U_A(S) - U_A(R_\lambda S)} = \frac{U_B(S) - U_B(R)}{U_A(S) - U_A(R)},$$

by the linearity of expected utility, i.e., $U(R_\lambda S) = \lambda U(R) + (1 - \lambda)U(S)$ for any vNM utility U . Moreover, as λ tends to zero, $U_i(S) - U_i(R_\lambda S)$ for $i = A, B$ tend to zero. Hence, S and $R_\lambda S$ generate a reversal for all λ small enough.

□

B Proof of Proposition 3

Proposition 3 shows how to construct the reversal-generating lotteries of Proposition 1 by generalizing Proposition 2 to arbitrarily different noise structures. The CARA case relies on the following lemma about diminishing utility differences as background wealth increases.

Lemma 4. *Let $S \geq_c R$. Then*

$$\lim_{x \rightarrow \infty} \text{CARA}_a(S+x) - \text{CARA}_a(R+x) = 0$$

for all $a > 0$.

Proof. Since S and R have equal means, we suppose, without loss of generality, that $\mathbb{E}[S] = \mathbb{E}[R] = 0$. Recall that $\text{cara}_a(x) = \frac{1-e^{-ax}}{a}$ denotes the CARA Bernoulli utility under coefficient a , and let m denote the essential infimum of S .²⁸ By monotonicity, $\text{cara}_a(x) - \text{cara}_a(m+x) \geq \text{cara}_a(x) - \text{CARA}_a(S+x)$ and by concavity of u_a , since $\mathbb{E}[S] = 0$, $\text{cara}_a(x) - \text{CARA}_a(S+x) \geq 0$. Moreover, since $m \leq 0$ and cara_a is concave,

$$\frac{d}{dx} \text{cara}_a(m+x) \cdot |m| \geq \text{cara}_a(x) - \text{cara}_a(m+x).$$

Note that $\frac{d}{dx} \text{cara}_a(m+x) = e^{-a(m+x)}$. Thus

$$e^{-a(m+x)} \cdot |m| \geq \text{cara}_a(x) - \text{cara}_a(m+x) \geq \text{cara}_a(x) - \text{CARA}_a(S+x) \geq 0.$$

Taking the limit as $x \rightarrow \infty$, we see that $\text{cara}_a(x) - \text{CARA}_a(S+x) \rightarrow 0$. Since S was an arbitrary mean-zero lottery and $\mathbb{E}[R] = 0$, we have

$$\lim_{x \rightarrow \infty} \text{cara}_a(x) - \text{CARA}_a(R+x) = 0$$

as well, concluding the proof. □

Proof of Proposition 3. For the case of CARA utilities, Lemma 1 establishes that for any pair of lotteries $S >_c R$, the ratio

$$\frac{U_B(S+x) - U_B(R+x)}{U_A(S+x) - U_A(R+x)}$$

²⁸This is the largest value that S exceeds with probability 1.

tends to infinity with x , while Lemma 4 establishes that the numerator and denominator go to zero. Hence, the result follows from Lemma 3.

For the CRRA case, Lemma 1 establishes that for any pair of lotteries $S >_c R$, the ratio

$$\frac{U_B(k \cdot S) - U_B(k \cdot R)}{U_A(k \cdot S) - U_A(k \cdot R)}$$

tends to infinity with k . As in the proof of Lemma 2,

$$\frac{U_B(k \cdot S) - U_B(k \cdot R_\lambda S)}{U_A(k \cdot S) - U_A(k \cdot R_\lambda S)} = \frac{U_B(k \cdot S) - U_B(k \cdot R)}{U_A(k \cdot S) - U_A(k \cdot R)},$$

and as λ tends to zero, $U_i(k \cdot S) - U_i(k \cdot R_\lambda S)$ tends to zero for $i = A, B$, and the result follows from Lemma 3. $\square$

C Proof of Theorem 1

Theorem 1 provides a characterization of vNM utilities that lead to consistent predictions under some noise structures. The proof of Theorem 1 makes use of Proposition 4 along with the following two lemmas.

Recall that in a WEU model (U, F, σ) , the probability of choosing X over Y is given by

$$F\left(\frac{U(X) - U(Y)}{\sigma(X, Y)}\right).$$

Let $f(t) = \frac{d}{dt}F(t)$. Note that f is continuous since $F \in \mathcal{F}$. In the following lemma, we establish compact convergence of $(\frac{1}{\sigma_n}f(\frac{t}{\sigma_n}))_n$ when $\sigma_n \rightarrow \sigma$. For notational convenience, we let

$$f_\sigma(t) = \frac{1}{\sigma}f\left(\frac{t}{\sigma}\right).$$

Lemma 5. *If $\sigma_n \rightarrow \sigma$, then $(f_{\sigma_n})_n$ converges compactly to f_σ .*

Proof of Lemma 5. Let $K \subset \mathbb{R}$ be compact.

$$\begin{aligned} & \sup_{t \in K} |f_{\sigma_n}(t) - f_\sigma(t)| \\ &= \sup_{t \in K} \left| \frac{1}{\sigma_n}f\left(\frac{t}{\sigma_n}\right) - \frac{1}{\sigma}f\left(\frac{t}{\sigma}\right) \right| \\ &= \sup_{t \in K} \left| \frac{1}{\sigma_n}f\left(\frac{t}{\sigma_n}\right) - \frac{1}{\sigma}f\left(\frac{t}{\sigma_n}\right) + \frac{1}{\sigma}f\left(\frac{t}{\sigma_n}\right) - \frac{1}{\sigma}f\left(\frac{t}{\sigma}\right) \right| \\ &\leq \left| \frac{1}{\sigma_n} - \frac{1}{\sigma} \right| \sup_{t \in K} \left| f\left(\frac{t}{\sigma_n}\right) \right| + \frac{1}{\sigma} \sup_{t \in K} \left| f\left(\frac{t}{\sigma_n}\right) - f\left(\frac{t}{\sigma}\right) \right| \end{aligned}$$

Note that since $\sigma_n \rightarrow \sigma$, there is a compact set S such that $\frac{t}{\sigma_n} \in S$ for all n high enough and $t \in K$. Since f is bounded on S , the first term tends to 0, and since $\frac{t}{\sigma_n}$ converges to $\frac{t}{\sigma}$ uniformly on K , and f is uniformly continuous on S , the second term tends to 0 as well. $\square$

Lemma 6. *Let $F_A, F_B \in \mathcal{F}$ and σ_A and σ_B be continuous. Let U_A and U_B be vNM utility functions with U_A more risk-averse. Suppose that for each $M \in \mathbb{R}$ there exist lotteries $S >_c R$ such that*

$$\frac{U_B(S) - U_B(R)}{U_A(S) - U_A(R)} \geq M.$$

Then there are reversals for (U_A, F_A, σ_A) and (U_B, F_B, σ_B) .

Proof of Lemma 6. Let U_A and U_B be as in the statement of the lemma. Let $Z, Z', \Lambda \in \mathcal{L}$ such that Z and Z' have the same distribution, and Λ is uniformly distributed on $[0, 1]$ and independent of Z and Z' . Let $F^A(t) = F_A\left(\frac{t}{\sigma_A(Z, Z')}\right)$ and $F^B(t) = F_B\left(\frac{t}{\sigma_B(Z, Z')}\right)$ and let $f^A(t) = \frac{d}{dt}F^A(t)$ and $f^B(t) = \frac{d}{dt}F^B(t)$. Let $\varepsilon_A > f^A(0)$ and $0 < \varepsilon_B < f^B(0)$, which is possible since $f^B(0)$ is positive. Since f^A and f^B are continuous at 0, there is $\delta > 0$ such that for all $0 < t < \delta$, $f^A(t) < \varepsilon_A$ and $f^B(t) > \varepsilon_B$.

By assumption, there are lotteries $X >_c Y$ such that

$$\frac{U_B(X) - U_B(Y)}{U_A(X) - U_A(Y)} \geq \frac{\varepsilon_A}{\varepsilon_B}.$$

Since the inequality only depends on the distributions of X and Y , we may choose X and Y to be independent of Λ . We define the random variables S_λ and R_λ by

$$S_\lambda(\omega) = \begin{cases} X(\omega) & \Lambda(\omega) \leq \lambda \\ Z(\omega) & \Lambda(\omega) > \lambda \end{cases}$$

and

$$R_\lambda(\omega) = \begin{cases} Y(\omega) & \Lambda(\omega) \leq \lambda \\ Z'(\omega) & \Lambda(\omega) > \lambda. \end{cases}$$

Note that for $\lambda \in (0, 1)$, $S_\lambda >_c R_\lambda$ and

$$U_A(S_\lambda) - U_A(R_\lambda) = \lambda(U_A(X) - U_A(Y)),$$

since $U_A(Z) = U_A(Z')$. Hence,

$$\frac{U_B(S_\lambda) - U_B(R_\lambda)}{U_A(S_\lambda) - U_A(R_\lambda)} = \frac{U_B(X) - U_B(Y)}{U_A(X) - U_A(Y)} \geq \frac{\varepsilon_A}{\varepsilon_B}, \quad (15)$$

and $U_A(S_\lambda) - U_A(R_\lambda), U_B(S_\lambda) - U_B(R_\lambda) > 0$.

Let $F_\lambda^A(t) = F_A\left(\frac{t}{\sigma_A(S_\lambda, R_\lambda)}\right)$, $F_\lambda^B(t) = F_B\left(\frac{t}{\sigma_B(S_\lambda, R_\lambda)}\right)$ and let $f_\lambda^A(t) = \frac{d}{dt}F_\lambda^A(t)$ and $f_\lambda^B(t) = \frac{d}{dt}F_\lambda^B(t)$. Since $S_\lambda \rightarrow Z$ and $R_\lambda \rightarrow Z'$ as $\lambda \rightarrow 0$, continuity of σ_A and σ_B , i.e., $\sigma_A(S_\lambda, R_\lambda) \rightarrow \sigma_A(S, R)$ and $\sigma_B(S_\lambda, R_\lambda) \rightarrow \sigma_B(S, R)$, imply that $f_\lambda^A \rightarrow f^A$ and $f_\lambda^B \rightarrow f^B$ compactly (Lemma 5). Thus, there is $\delta' > 0$ and λ_A such that for all $\lambda \in (0, \lambda_A)$ and all $t \in [0, \delta']$, $f_\lambda^A(t) < \varepsilon_A$ and $F_\lambda^A(t) < \frac{1}{2} + t\varepsilon_A$. Likewise, there is λ_B such that for all $\lambda \in (0, \lambda_B)$ and all $t \in [0, \delta']$, $F_\lambda^B(t) > \frac{1}{2} + t\varepsilon_B$.

For λ small enough,

$$U_A(S_\lambda) - U_A(R_\lambda) < \delta' \quad \text{and} \quad U_B(S_\lambda) - U_B(R_\lambda) < \delta'.$$

Thus, for λ small enough, we have

$$\begin{aligned} F_\lambda^B(U_B(S_\lambda) - U_B(R_\lambda)) &> \frac{1}{2} + (U_B(S_\lambda) - U_B(R_\lambda))\varepsilon_B \\ &\geq \frac{1}{2} + (U_A(S_\lambda) - U_A(R_\lambda))\varepsilon_A \\ &> F_\lambda^A(U_A(S_\lambda) - U_A(R_\lambda)). \end{aligned}$$

The second inequality follows from (15). □

Proof of Theorem 1. Let U_A be more risk-averse than U_B . In Section 4, we showed that if $U_A \geq_{\text{rel}} U_B$ then there are continuous (indeed, constant) σ_A and σ_B and $F \in \mathcal{F}$ such that there are no reversals under (U_A, F, σ_A) and (U_B, F, σ_B) . We now show that when there will always be reversals if U_A is not relatively more concave than U_B . Indeed, suppose that, for all $k > 0$, it is not the case that $kU_A - U_B$ is concave, i.e., kU_A is not absolutely more concave than U_B . Then, by Proposition 4, for each $k > 0$, there are lotteries $S \succ_c R$ such that

$$k(U_A(S) - U_A(R)) < U_B(S) - U_B(R).$$

Hence, the result follows from Lemma 6. □

D Comparison with Standard Comparative Risk Notions

Note that U_A is more risk-averse than U_B in the Arrow-Pratt sense if and only if, for all x ,

$$\frac{u_B''(x)}{u_A''(x)} \leq \frac{u_B'(x)}{u_A'(x)}.$$

While stochastic comparative risk depends on bounding the ratio of second derivatives by a constant, the traditional Arrow-Pratt notion of comparative risk depends on bounding this ratio by the ratio of marginal utilities. Thus, when $u_B'(x)/u_A'(x)$ is bounded, the traditional notion of comparative risk is sufficient for the stochastic notion. On the other hand, Ross' (1981) stronger comparative notion requires

$$\frac{u_B''(x)}{u_A''(x)} \leq k \leq \frac{u_B'(x)}{u_A'(x)}, \quad (16)$$

for some $k > 0$ and all x , implying the boundedness of u_B''/u_A'' , i.e., $u_A \geq_{\text{rel}} u_B$. Moreover, Ross (1981) shows (16) is equivalent to the existence of k such that $ku_A - u_B$ is concave and decreasing, while $u_A \geq_{\text{rel}} u_B$ does not require $ku_A - u_B$ to be decreasing.

The next proposition states that, in the presence of noise related by a scale factor, Ross' (1981) notion captures stochastic comparative risk aversion stemming from preferences rather than greater choice precision. To formalize this, we introduce the following notion of relative choice precision.

We say that A has *greater choice precision* than B if, whenever X first-order dominates Y , A chooses X more than B .

Proposition 7. *Let $U_A = \mathbb{E}[u_A]$ and $U_B = \mathbb{E}[u_B]$ be vNM utilities with U_A more risk-averse than U_B . Define $F_A(t) = F(t/\sigma_A)$ and $F_B(t) = F(t/\sigma_B)$ for some $\sigma_A, \sigma_B > 0$ and $F \in \mathcal{F}$. Then there are no reversals for (U_A, F_A) and (U_B, F_B) , and B has greater choice precision if and only if $u_A/\sigma_A - u_B/\sigma_B$ is decreasing and concave.*

Proof of Proposition 7. For any $X, Y \in \mathcal{L}$, A chooses X over Y more than B if and only if

$$U_A(X)/\sigma_A - U_B(X)/\sigma_B \geq U_A(Y)/\sigma_A - U_B(Y)/\sigma_B. \quad (17)$$

Thus, if $u_A/\sigma_A - u_B/\sigma_B$ is decreasing and concave, then A chooses X less often when X first-order dominates Y and more often when X concave-order dominates Y .

Conversely, suppose that there are no reversals and B has greater choice precision. It follows from Proposition 4 that $u_A/\sigma_A - u_B/\sigma_B$ is concave. Moreover, for any $x > y$, a degenerate lottery X paying x first-order dominates Y paying y . Since B has greater choice precision, the right-hand side of (17) is larger, and we conclude that $u_A/\sigma_A - u_B/\sigma_B$ is decreasing. □

E Expo-power Utilities

We illustrate the expo-power utility function, a two-parameter family proposed by Saha (1993) to capture increasing/decreasing absolute/relative risk aversion,²⁹ suffers from the same problems as CARA and CRRA utilities. The increasing and concave expo-power utility function is given by

$$u_{a,r}(x) = \frac{1 - e^{-ax^{1-r}}}{a},$$

for $x \geq 0$ and positive a and $0 \leq r < 1$. Note that the absolute risk aversion is given by

$$A_{a,r}(x) = \frac{r}{x} + a(1-r)x^{-r},$$

for $x > 0$, which is decreasing, constant, or increasing depending on the parameters. Thus, unlike CARA or CRRA utility, the family of expo-power utilities, parameterized by a and r , is not totally ordered by absolute/relative risk aversion.

Importantly, for any pairs of coefficients (a_1, r_1) and (a_2, r_2) such that $A_{a_1, r_1}(x) \geq A_{a_2, r_2}(x)$ so that $U_1 = \mathbb{E}[u_{a_1, r_1}(X)]$ is more risk-averse than $U_2 = \mathbb{E}[u_{a_2, r_2}(X)]$, it must hold that $r_1 = r_2$ and $a_1 > a_2$. A simple calculation shows that when $r_1 = r_2$ and $a_1 > a_2$, $u''_{a_2, r_2}(x)/u''_{a_1, r_1}(x)$ tends to infinity as $x \rightarrow \infty$. Thus, we have the following corollary.

Corollary 3. *Let $F_A, F_B \in \mathcal{F}$ and σ_A, σ_B be continuous. For any distinct expo-power utility functions U_A and U_B there are reversals for (U_A, F_A, σ_A) and (U_B, F_B, σ_B) .*

Corollary 3 highlights that the paradoxical properties arising from CARA (CRRA) utilities are pervasive, and that they do not depend on the strong assumption of constant absolute (relative) risk aversion.

²⁹Notably, Holt and Laury (2002) use the expo-power utility function to model increasing relative risk aversion exhibited in their data.

F Sum-ex and Sum-power Utility Functions

In Section 5.1, we demonstrated that for $h > \ell \geq 0$ the utility families

$$u_a(x; h, \ell) = a \cdot \text{cara}_h(x) + \text{cara}_\ell(x) \quad a \geq 0$$

and

$$v_a(x; h, \ell) = a \cdot \text{crra}_h(x) + \text{crra}_\ell(x) \quad a \geq 0,$$

which are parametrized by the coefficient a placed on the more risk-averse cara or crra utility do not lead to reversals under homoskedastic Fechnerian noise. A utility function taking the form $u_a(x; h, \ell)$ is a sum-ex utility and one taking the form $v_a(x; h, \ell)$ is a sum-power utility. In this appendix, we briefly discuss the axioms that characterize the sum-ex and sum-power families as well as an interpretation of the risk parameter a .

F.1 One-switch Utilities

Bell (1988, 1996) showed that sum-ex and sum-power utilities are characterized by *one-switch* properties on preferences over gambles and decreasing absolute risk aversion.

Definition 4. *We say that a Bernoulli utility u satisfies the additive one-switch rule if, for all X and Y*

$$\mathbb{E}[u(X + t) - u(Y + t)]$$

is single-crossing in t . Likewise, we say that u satisfies the multiplicative one-switch rule if, for all X and Y

$$\mathbb{E}[u(kX) - u(kY)]$$

is single-crossing in $k > 0$.

The family of CARA utilities is similarly characterized by a zero-switch condition, which requires that preferences over gambles are invariant to background wealth. However, we often observe individuals who become less risk-averse as their wealth levels increase, exhibiting *decreasing absolute risk aversion* (DARA). Formally, we say that a Bernoulli utility function u exhibits DARA if $-u''(x)/u'(x)$ is decreasing in x .

As argued by Bell (1988), the additive one-switch property is normatively appealing under DARA, since one may start to prefer a riskier gamble once their wealth exceeds some point and their aversion to risk diminishes. However, this preference should not be reverted at even higher levels of background wealth, at which the individual is even more risk tolerant. Similarly, Bell (1996) argues that the multiplicative one-switch rule is a normatively appealing criterion for an individual comparing two investments with different stochastic returns.

The following proposition is due to Bell (1988, 1996) and Pedersen and Satchell (2001).

Proposition 8. *Suppose a concave infinitely differentiable Bernoulli utility function u satisfies decreasing absolute risk aversion. Then*

1. *u satisfies the additive one-switch rule if and only if $u(x)$ is a sum-ex utility function;*
2. *u satisfies the multiplicative one-switch rule if and only if $u(x)$ is a sum-power utility function.*

F.2 Interpreting the Risk Coefficient

In Section 5.1, we showed that for sum-ex and sum-power utilities

$$u_a(x; h, \ell) = a \cdot \text{cara}_h(x) + \text{cara}_\ell(x), \quad v_a(x; h, \ell) = a \cdot \text{crra}_h(x) + \text{crra}_\ell(x),$$

a higher coefficient a corresponds to greater risk aversion in the Arrow-Pratt sense and an increased likelihood of choosing safer lotteries under identical noise. Thus, we advocated for replacing CARA and CRRA utilities with sum-ex and sum-power utilities in estimating risk attitudes.

In this section, we further provide an interpretation of the risk coefficient a that is to be estimated for these families. Specifically, we show that the coefficient a can be interpreted as a background wealth level. That is, the preference of an individual with a lower coefficient a is identical to that of an individual with a larger coefficient and more background wealth. This equivalence is formalized in the next proposition.

Proposition 9. *Let $a > b > 0$ and $h > \ell > 0$. Then there are $\alpha, \beta, t, k > 0$, $\gamma, \delta \in \mathbb{R}$ such that*

$$u_b(x; h, \ell) = \alpha \cdot u_a(x + t; h, \ell) + \gamma \quad \text{for all } x \in \mathbb{R} \quad (18)$$

$$v_b(x; h, \ell) = \beta \cdot v_a(k \cdot x; h, \ell) + \delta \quad \text{for all } x > 0. \quad (19)$$

Proof of Proposition 9. For sum-ex utility, let

$$\alpha = \left(\frac{a}{b}\right)^{\frac{\ell}{h-\ell}}, \quad \gamma = \frac{b - \alpha \cdot a}{h} + \frac{1 - \alpha}{\ell}, \quad t = \frac{1}{h - \ell} \ln\left(\frac{a}{b}\right).$$

For sum-power utility, if $\ell \neq 1$ and $h \neq 1$ let

$$\beta = \left(\frac{a}{b}\right)^{\frac{\ell-1}{h-\ell}}, \quad \delta = \frac{a \cdot \beta - b}{1 - h} + \frac{\beta - 1}{1 - \ell}, \quad k = \beta^{\frac{1}{\ell-1}}.$$

Plugging in these parameters, we obtain (18) and (19). For h or ℓ equal to 1, taking the limits of the expressions for β , δ , and γ works as well, completing the proof. $\square$

Proposition 9 highlights a benefit that sum-ex and sum-power share with CARA and CRRA in the deterministic setting. One important feature of CARA utility is that background wealth does not affect preferences and thus does not need to be separately estimated. By Proposition 9, the estimation of the coefficient on the higher CARA utility can itself be interpreted as an estimation of background wealth which does not, therefore, need to be separately estimated. Likewise, CRRA is often used as a preference specification over investments where it has the benefit that the level of stakes to be invested do not need to be estimated since they do not affect the preference. While sum-power preferences are not invariant to changes in stakes, the coefficient on the higher CRRA utility can be interpreted as revealing the level of stakes to be invested, which do not then need to be separately estimated.

G APU Models

G.1 Homogeneous Costs

Proposition 10. *Let ρ_A have the APU representation (U_A, c) and ρ_B have the representation and (U_B, c) where U_A and U_B are vNM utilities with U_A more risk-averse. The following are equivalent:*

1. *There are no compound reversals;*
2. *There are no FOSD reversals;*
3. $U_A \geq_{\text{abs}} U_B$.

Proposition 10 generalizes Proposition 4 to the broader APU framework by establishing the equivalence between $\geq_{\text{abs}}$ and no reversals under a shared cost function c , extending the result beyond binary choices from concave ordered pairs.

Proof of Proposition 10. Since each FOSD reversal is a compound reversal, (1) implies (2). We now show that (3) implies (1).

Fix a menu M . Since c is strictly convex, ρ_A^M is the unique argmax of

$$\sum_{X \in M} U_A(X)p(X) - c(p(X)),$$

and ρ_B^M is the unique argmax of

$$\sum_{X \in M} U_B(X)p(X) - c(p(X)).$$

If $\rho_A^M = \rho_B^M$, then there is no compound reversal, so suppose they are distinct. We thus have the following two inequalities:

$$\begin{aligned} \sum_{X \in M} U_A(X)\rho_A^M(X) - c(\rho_A^M(X)) &> \sum_{X \in M} U_A(X)\rho_B^M(X) - c(\rho_B^M(X)), \\ \sum_{X \in M} U_B(X)\rho_B^M(X) - c(\rho_B^M(X)) &> \sum_{X \in M} U_B(X)\rho_A^M(X) - c(\rho_A^M(X)). \end{aligned}$$

Summing the inequalities gives

$$\sum_{X \in M} (U_A - U_B)(X)\rho_A^M(X) > \sum_{X \in M} (U_A - U_B)(X)\rho_B^M(X).$$

Since $U_A \geq_{\text{abs}} U_B$ by (3), $U_A - U_B = \mathbb{E}[v]$ for some concave function v , so we do not have a compound reversal.

Finally, we show that (2) implies (3). Contrapositively, suppose that $U_A - U_B = \mathbb{E}[v]$ for some function v that is not concave. Then there are lotteries $S >_c R$ such that $\mathbb{E}[v(S)] < \mathbb{E}[v(R)]$. Let $M = \{S, R\}$.

The maximization of APU yields the first-order conditions:

$$\begin{aligned} U_A(S) - U_A(R) &= c'(\rho_A^M(S)) - c'(1 - \rho_A^M(S)) \\ U_B(S) - U_B(R) &= c'(\rho_B^M(S)) - c'(1 - \rho_B^M(S)). \end{aligned}$$

Let $g(t) = c'(t) - c'(1-t)$. Since c is strictly convex, c' is strictly increasing, so g is strictly increasing.³⁰ Subtracting the first-order condition for B from that of A yields

$$\mathbb{E}[v(S) - v(R)] = g(\rho_A^M(S)) - g(\rho_B^M(S)).$$

Since $\mathbb{E}[v(S)] < \mathbb{E}[v(R)]$, we conclude that $\rho_A^M(S) < \rho_B^M(S)$, which is an FOSD reversal. $\square$

G.2 Heterogeneous Costs

In this section, we show that the APU representation (U, c) for ρ reduces to the FEU model of Section 3 on binary menus under a smoothness assumption on the cost function. Recall that a cost function is a function $c: [0, 1] \rightarrow \mathbb{R} \cup \{\infty\}$ that is strictly convex and continuously differentiable on $(0, 1)$ and satisfies $\lim_{p \rightarrow 0} c'(p) = -\infty$. Also, recall that $\mathcal{F}$ is the set of all C_+^1 functions $F: \mathbb{R} \rightarrow [0, 1]$ satisfying $F(t) + F(-t) = 1$.

Lemma 7. *Let ρ have the APU representation (U, c) . If $c \in C_+^2$, then for all binary menus,*

$$\rho(X, Y) = F(U(X) - U(Y))$$

for some $F \in \mathcal{F}$.

Proof of Lemma 7. Fix an APU model (U, c) and define $g(p) = c'(p) - c'(1-p)$. Suppose that the restriction of c to $(0, 1)$ is a C_+^2 function. Then $g \in C_+^2$. Moreover, since g must satisfy $g(1-p) = -g(p)$ and $\lim_{p \rightarrow 0} c'(p) = -\infty$, we have $\lim_{p \rightarrow 0} g(p) = -\infty$ and $\lim_{p \rightarrow 1} g(p) = \infty$. Let $F: \mathbb{R} \rightarrow (0, 1)$ be the inverse of g . By the inverse function theorem, $F \in C_+^1$. Moreover, we have $1 - F(t) = F(-t)$, so $F \in \mathcal{F}$.

Finally, we show that the Fechnerian model (U, F) and the APU model (U, c) coincide for binary menus. Fix a binary menu $\{X, Y\}$ and let p be the choice probability of X under the APU model (U, c) . The maximization of the APU model yields the first-order condition:

$$U(X) - U(Y) = c'(p) - c'(1-p) = g(p).$$

Applying F to the first and last terms, we obtain $F(U(X) - U(Y)) = p$. $\square$

³⁰Since c is strictly convex, $c(y) < c(x) + c'(y)(y-x)$ and $c(x) < c(y) + c'(x)(x-y)$ for $x < y$, which yield $(c'(y) - c'(x))(y-x) > 0$. Thus c' is strictly increasing.

Proof of Theorem 2. First, we show (3) implies (1). Let $k > 0$ such that $kU_A \geq_{\text{abs}} U_B$ and let c be any cost function. Let ρ_A and ρ_B have the APU representations (kU_A, c) and (U_B, c) , respectively. By Proposition 10, ρ_A and ρ_B do not yield compound reversals. Since $(U_A, \frac{c}{k})$ represents ρ_A and (U_B, c) represents ρ_B , we have (3) implies (1) by setting $c_A = c/k$ and $c_B = c$.

Since each FOSD reversal is a compound reversal, (1) implies (2). We conclude by showing that (2) implies (3). Contrapositively, consider U_A that is not relatively more concave than U_B . Then by Lemma 7, it follows from Theorem 1 that, for any cost functions $c_A, c_B \in C_+^2$, there exists a binary menu $\{S, R\}$ where $S >_c R$ that generates a reversal, which is an FOSD reversal. $\square$

H Random Parameter Models

H.1 Compound Reversals for CARA- and CRRA-based RPMs

In this appendix, we construct examples of compound reversals in CARA- and CRRA-based random parameter models.

Let $p = 10^{-5}$. Consider the menu $M = \{X, Y, Z\}$ with the following distributions:

$$X = \begin{cases} 20 & \text{w.p. } 1-p \\ 10 & \text{w.p. } p \end{cases}, \quad Y = \begin{cases} 23 & \text{w.p. } \frac{1-p}{2} \\ 20 & \text{w.p. } \frac{p}{2} \\ 19 & \text{w.p. } \frac{1-p}{2} \\ 10 & \text{w.p. } \frac{p}{2} \end{cases}, \quad Z = \begin{cases} 27 & \text{w.p. } \frac{1-p}{2} \\ 20 & \text{w.p. } p \\ 17 & \text{w.p. } \frac{1-p}{2} \end{cases}.$$

For CARA random parameter models, let ε be uniformly distributed on $[0, 0.5]$. Let A and B have absolute risk coefficients 1.16 and 0.1, respectively. Under the random parameter model with ε added to risk coefficients, we have

$$\rho_A^M(X) = \rho_A^M(Z) > 0 \quad \text{and} \quad \rho_B^M(Y) = 1.$$

Since $Y >_c X_{\frac{1}{2}}Z$, $Y >_c W$ for any W that is a mixture of X, Y, Z with positive equal weights on X and Z . Thus, we have a compound reversal. Similarly, we have a compound reversal of CRRA random parameter models from the same menu, when A and B have relative risk coefficients 18.77 and 2.2, respectively and ε is uniformly distributed on $[0, 12]$.

H.2 Proof of Theorem 3

Proof of Theorem 3. We begin by showing that random parameter models on a collinear family $\mathcal{U} = \{U_a\}_{a \in \mathcal{A}}$ cannot lead to compound reversals. To this end, we first note that for any lotteries X, Y and parameters $a > b > c$, if $(U_b(X) - U_b(Y))(U_c(X) - U_c(Y)) < 0$, then $(U_a(X) - U_a(Y))(U_b(X) - U_b(Y)) > 0$. That is, if U_b and U_c rank X and Y differently, U_a 's ranking coincides with U_b 's.³¹ To see this, consider the transformed utilities V_i that lie on a line: $V_b = \alpha V_a + (1 - \alpha)V_c$, so that $V_b(X) - V_b(Y) = \alpha(V_a(X) - V_a(Y)) + (1 - \alpha)(V_c(X) - V_c(Y))$.

For each menu M , we can thus find $\bar{a} \in \mathcal{A}$ such that for all $a \geq \bar{a}$,

$$(U_a(X) - U_a(Y))(U_{\bar{a}}(X) - U_{\bar{a}}(Y)) \geq 0$$

for all $X, Y \in M$.

Let ρ_A, ρ_B be random parameter models on $\mathcal{U}$ with corresponding parameter distributions $\mu_A, \mu_B \in \Delta(\mathcal{A})$ and suppose that A is more risk-averse, meaning $\mu_A >_{\text{FOSD}} \mu_B$. There thus exist random variables ξ_A, ξ_B defined on some probability space $(\Theta, \mathcal{F}, \eta)$ with corresponding laws μ_A and μ_B such that $\xi_A(\theta) \geq \xi_B(\theta)$ for $\theta \in \Theta$. For each θ , let $X_\theta = \operatorname{argmax}_{X \in M} U_{\xi_A(\theta)}(X)$ and $Y_\theta = \operatorname{argmax}_{X \in M} U_{\xi_B(\theta)}(X)$. Since $\xi_A(\theta) \geq \xi_B(\theta)$, $U_{\bar{a}}(X_\theta) \geq U_{\bar{a}}(Y_\theta)$. Note that $\rho_A^M(X) = \eta(X_\theta = X)$ and $\rho_B^M(X) = \eta(Y_\theta = X)$. It follows that

$$\sum_{X \in M} U_{\bar{a}}(X) \rho_A^M(X) \geq \sum_{X \in M} U_{\bar{a}}(X) \rho_B^M(X).$$

Moreover, this inequality is strict unless $X_\theta = Y_\theta$ almost surely, in which case $\rho_A^M = \rho_B^M$. Thus, since $U_{\bar{a}}$ is risk-averse, $U_{\bar{a}} = \mathbb{E}[u]$ for some concave function and there are no compound reversals.

We next show that when there are no compound reversals, $\mathcal{U}$ is collinear. We define a linear system and show that any solution enables us to construct a compound reversal. Let $x_0 < x_1 < x_2 < x_3 \in \mathbb{R}$ and let $\alpha_0, \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$. Let $a < b < c \in \mathcal{A}$ with corresponding vNM utilities $U_a = \mathbb{E}[u_a], U_b = \mathbb{E}[u_b], U_c = \mathbb{E}[u_c]$, where, without loss of generality, we normalize $u_k(x_0) = 0$ and $u_k(x_1) = 1$ for $k = a, b, c$.

³¹In fact, such a condition is used to characterize collinear families by Kartik, Lee, and Rappoport (2024).

Suppose that there are $\alpha_0, \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ such that

$$\sum_{i=0}^3 \alpha_i u_a(x_i) = 0, \quad \sum_{i=0}^3 \alpha_i u_b(x_i) > 0, \quad \sum_{i=0}^3 \alpha_i u_c(x_i) < 0. \quad (20)$$

Note that the value of each $\sum_{i=0}^3 \alpha_i u_k(x_i)$ is independent of α_0 , so we choose α_0 such that $\sum_i \alpha_i = 0$. Moreover, since the sign $\sum_{i=0}^3 \alpha_i u_k(x_i)$ is invariant to positive scaling, it suffices to consider α such that $\sum_i |\alpha_i| = 2$. It is easy to see that $\sum_{i=0}^3 \alpha_i u_k(x_i) = U_k(X) - U_k(Y)$ for some lotteries X and Y and for all k . Let $t = \frac{1}{2}\mathbb{E}[X + Y]$ and consider a menu $M = \{X, Y, Z\}$, where Z is distributed as $t_\varepsilon(X_{\frac{1}{2}}Y)$. Note that $Z >_c X_{\frac{1}{2}}Y$.

Since U_a is risk-averse and indifferent between X and Y , it is uniquely maximized at Z . For ε small enough, X is the unique maximizer of U_b and Y is the unique maximizer of U_c . The distributions $\mu_A = \frac{1}{2}\delta_b + \frac{1}{2}\delta_c$ and $\mu_B = \delta_a$ correspond to A being more risk than B . However, the compound lottery induced by A 's choices (which is distributed as $X_{\frac{1}{2}}Y$) is dominated by the compound lottery induced by B 's choices (which is simply Z), yielding a compound reversal.

Thus, for there not to exist any compound reversals between random parameter models on $\mathcal{U}$, it must be that there is no solution to (20). By Motzkin's transposition theorem, there exist $\beta_0, \beta_1, \beta_2 \in \mathbb{R}$ where $\beta_1, \beta_2 \geq 0$, with at least one positive such that $\beta_1 u_b(x_i) = \beta_0 u_a(x_i) + \beta_2 u_c(x_i)$ for $i = 0, 1, 2, 3$. Since x_2 and x_3 were arbitrary, we conclude that $\beta_1 u_b(x) = \beta_0 u_a(x) + \beta_2 u_c(x)$ for all $x \in \mathbb{R}$. Since U_c is more risk-averse than U_b which is more risk-averse than U_a which is itself risk-averse, we conclude that each β_i is non-zero. Thus,

$$u_b = \frac{\beta_0}{\beta_1} u_a + \frac{\beta_2}{\beta_1} u_c.$$

Since $u_k(x_1) = 1$ for all k , $\frac{\beta_0}{\beta_1} + \frac{\beta_2}{\beta_1} = 1$, and we conclude that the family $\mathcal{U}$ is collinear. $\square$